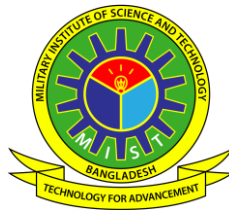


THERMOMECHANICAL BEHAVIOR OF FIBER- REINFORCED COMPOSITE MATERIALS

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B.Sc. ENGINEERING THESIS



**DEPARTMENT OF AERONAUTICAL ENGINEERING
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DHAKA, BANGLADESH**

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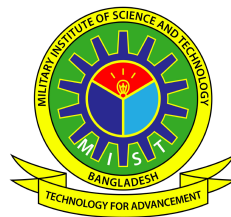
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A Thesis Submitted in Partial Fulfillment of the Requirements for the
Degree of Bachelor of Science in Aeronautical Engineering



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DECLARATION

We hereby declare that the study reported in this thesis entitled as above is our own original work and has not been submitted before anywhere for any degree or other purposes. Further we certify that the intellectual content of this disease is the product of our own work and that all the assistance received in preparing this thesis and sources have been acknowledged and cited in the reference section.

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ABSTRACT

Thermomechanical Behavior of Fiber-Reinforced Composite Materials

Fiber-reinforced composite materials are increasingly used in aerospace and structural applications due to their high strength-to-weight ratio and directional stiffness. When such structures are subjected to thermal loading, both temperature and stress fields develop simultaneously, making the analysis more complex than a purely mechanical problem. This study presents a finite difference solution for the steady-state thermomechanical behaviour of an orthotropic Al–B₄C metal matrix composite plate. The material properties are determined using the Rule of Mixtures, and the governing thermoelastic equations are reduced to a single fourth-order equation in a displacement potential function, which is then solved numerically on a uniform grid. The effects of applied temperature difference and edge boundary conditions on the induced stress distributions are examined. The results show that the lateral normal stress is the dominant stress component over most of the plate, while shear stress concentrates near the constrained edges. The numerical results are validated against finite element solutions obtained from ABAQUS, and the agreement is found to be satisfactory over most of the domain. The study demonstrates that the present finite difference scheme can be used effectively for the stress analysis of orthotropic composite plates under thermal loading.

Thermomechanical Behavior of Fiber Reinforced Composite Materials

ফাইবার-রিইনফোর্সড কম্পোজিট উপকরণ তাদের উচ্চ শক্তি-অনুপাত এবং দিকনির্ভর দৃঢ়তার কারণে কাঠামোগত প্রকৌশলে ব্যাপকভাবে ব্যবহৃত হচ্ছে। তাপীয় লোডের অধীনে এই ধরনের কাঠামো বিশ্লেষণ করা জটিল, কারণ তাপমাত্রা ও পীড়ন ক্ষেত্র একসাথে তৈরি হয়। এই গবেষণায় একটি অর্থোট্রপিক $Al - B_4C$ মেটাল ম্যাট্রিক্স কম্পোজিট প্লেটের স্থিতিশীল তাপ-যান্ত্রিক আচরণ ফিনিট ডিফারেন্স পদ্ধতিতে বিশ্লেষণ করা হয়েছে। উপাদানের বৈশিষ্ট্যগুলো রুল অব মিক্সচার ব্যবহার করে নির্ধারণ করা হয়েছে। তাপস্থিতিস্থাপক সমীকরণগুলোকে একটি ডিসপ্লেসমেন্ট পটেনশিয়াল ফাংশনের সাপেক্ষে একটি একক চতুর্থ ক্রমের আংশিক ডিফারেনশিয়াল সমীকরণে রূপান্তর করা হয়েছে এবং সেটি সমান ব্যবধানের গ্রিডে সংখ্যাগতভাবে সমাধান করা হয়েছে। দুটি ভিন্ন বাউন্ডারি কন্ডিশন সহ দুটি সমস্যা বিশ্লেষণ করা হয়েছে। ফলাফলে দেখা গেছে যে প্লেটের বেশিরভাগ অংশজুড়ে পার্শ্বিক নরমাল পীড়নই প্রধান, এবং শিয়ার পীড়ন সীমানার কাছে কেন্দ্রীভূত হয়। প্রয়োগকৃত তাপমাত্রার পার্থক্যের সাথে সমস্ত পীড়ন উপাদানগুলো সমানুপাতিকভাবে পরিবর্তিত হয়, যা রৈখিক তাপস্থিতিস্থাপক আচরণের সাথে সামঞ্জস্যপূর্ণ। ফিনিট ডিফারেন্স ফলাফল ফিনিট এলিমেন্ট সমাধানের (FEM) সাথে তুলনা করা হয়েছে এবং ডোমেনের অধিকাংশ অঞ্চলে সন্তোষজনক মিল পাওয়া গেছে।

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LIST OF NOTATION

E_1, E_2	Young's moduli in fiber and transverse material directions
G_{12}	In-plane shear modulus of orthotropic lamina
g	Volumetric internal heat generation rate (W m^{-3})
h	Finite difference mesh spacing in x -direction
h_c	Convective heat transfer coefficient
k	Finite difference mesh spacing in y -direction
k_1, k_2	Thermal conductivity in fiber (x) and transverse (y) directions ($\text{W m}^{-1}\text{K}^{-1}$)
L_x, L_y	Panel dimensions in x - and y -directions
N_x, N_y	Number of finite difference grid nodes in x - and y -directions
$Q_{11}, Q_{12}, Q_{22}, Q_{66}$	Reduced stiffness coefficients in principal material axes Jones (1999)
T	Temperature field
T_0	Ambient (reference) temperature
u_x, u_y	In-plane displacement components in x - and y -directions
V_f, V_m	Fiber and matrix volume fractions
w	Panel thickness (out-of-plane dimension)
α_1, α_2	Coefficients of thermal expansion (CTE) in fiber and transverse directions
$\beta_1, \dots, \beta_6$	Ansatz constants in displacement function formulation
ΔT	Temperature change from stress-free reference state
$\varepsilon_x, \varepsilon_y, \gamma_{xy}$	In-plane normal and shear strain components
ν_{12}, ν_{21}	Poisson's ratios (major and minor) of orthotropic lamina
ψ	Scalar displacement potential function
$\sigma_{xx}, \sigma_{yy}, \tau_{xy}$	In-plane normal and shear stress components
ABAQUS	Commercial finite element analysis software
CTE	Coefficient of Thermal Expansion
FDM	Finite Difference Method
FEA	Finite Element Analysis
FEM	Finite Element Method
FRPC	Fiber-Reinforced Polymer Composite
MATLAB	Matrix Laboratory (numerical computing environment)
MMC	Metal Matrix Composite
PDE	Partial Differential Equation

CHAPTER 1

INTRODUCTION

1.1 Overview

Fiber-reinforced composite materials, their directional stiffness, and the thermomechanical response under thermal loading are covered in this chapter. The background and motivation for the present study are discussed, followed by an outline of the thesis organisation.

1.2 Introduction

A composite material consists of two or more constituent materials combined at a macroscopic scale to produce properties that neither constituent possesses individually. In a fiber-reinforced composite (FRC), stiff and strong fibers are embedded in a continuous matrix phase that binds the fibers together, transfers load between them, and protects them from the environment. The fibers carry the majority of the applied load, while the matrix provides lateral support and distributes stress among the fibers (Gibson, 2016; Kaw, 2005).

As shown in Figure 1.1, the directional arrangement of fibers within the matrix gives the lamina its characteristic anisotropic properties.

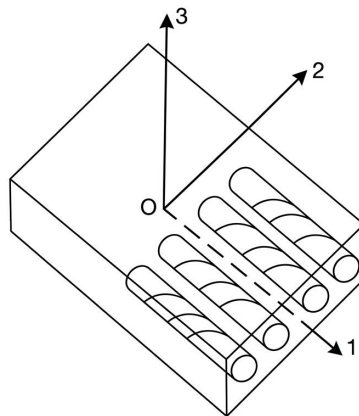


Figure 1.1: Schematic of a fiber-reinforced composite lamina showing the fiber phase embedded in the matrix phase.

The combination of a light, tough matrix with high-stiffness fibers yields a material with a very high strength-to-weight and stiffness-to-weight ratio compared to conventional metals. This makes fiber-reinforced composites attractive for aerospace, automotive, and civil structural applications, where minimizing structural mass is a primary design driver (Soutis, 2005).

Unlike metals, whose properties are the same in every direction (isotropic), a fiber-reinforced lamina has directional properties determined by the fiber orientation. When fibers are aligned along a single direction, the lamina is stiffer and stronger along the fiber axis than across it. Such a material is said to be *orthotropic*: its properties are symmetric about three mutually perpendicular planes, called the principal material axes (Jones, 1999). Figure 1.2 illustrates the principal material axes relative to the global plate coordinate system.

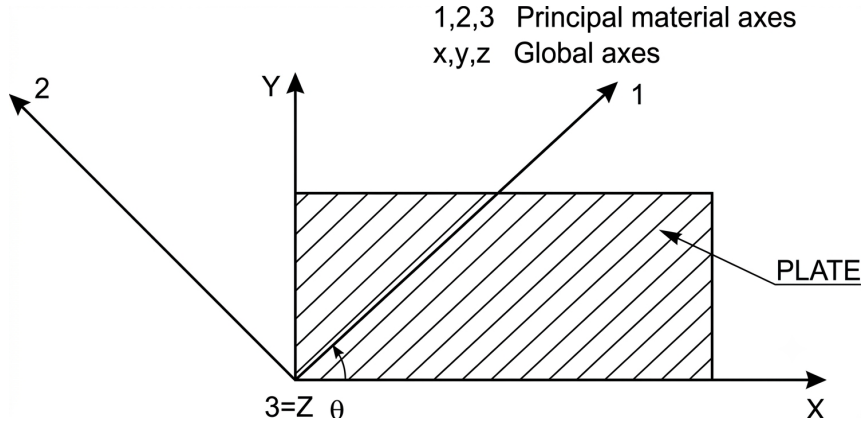


Figure 1.2: Principal material axes (1–2) of a unidirectional fiber-reinforced lamina and the plate coordinate axes (x – y) for fiber orientation angle θ . When $\theta = 0^\circ$, the fiber axis is aligned with the x -axis, which is the configuration adopted throughout this study.

The principal material directions are labelled 1 and 2, where direction 1 runs along the fibers and direction 2 runs transverse to them. The elastic modulus along the fiber direction (E_1) is typically several times larger than the transverse modulus (E_2). In the present work the fibers are aligned with the global x -axis ($\theta = 0^\circ$), so the principal material axes coincide with the coordinate axes throughout.

The relative proportion of fibers is quantified by the fiber volume fraction V_f , defined as the volume of fibers divided by the total composite volume. The matrix volume fraction is $V_m = 1 - V_f$. In the present work V_f is varied from 0.3 to 0.7 to examine its influence on the thermomechanical response (Gibson, 2016).

1.3 Background And Motivation

Fiber-reinforced composite materials are increasingly deployed in high-temperature structural applications in aerospace and power-generation industries, where components are routinely subjected to severe and non-uniform thermal environments (Kaw, 2005; Soutis, 2005). In these scenarios, the orthotropic thermal expansion characteristics of the composite lamina couple the temperature field directly to the mechanical stress field: differential thermal expansions restrained by boundary conditions or by the directional stiffness of the material generate internal stresses that can be substantial even in the absence of mechanical loading (Noda et al., 2003).

A metal matrix composite (MMC) uses a metal alloy as the matrix phase rather than a polymer resin. Aluminium-based MMCs are particularly common in aerospace structures because they combine the low density and corrosion resistance of aluminium with the high stiffness of ceramic or carbon fibers (Kaw, 2005). Despite their widespread use, the thermomechanical stress state in orthotropic MMC plates under realistic boundary conditions is not straightforward to analyse. Closed-form solutions exist only for simple loading and geometry configurations; for general boundary conditions a numerical approach is required.

Figure 1.3 illustrates the physical scenario addressed in this work: a rectangular composite plate subjected to an edge temperature differential, where the resulting thermal gradient induces a constrained stress field throughout the domain.

The finite difference method (FDM) offers a transparent and computationally efficient route to solving the coupled thermoelastic equations on a uniform rectangular grid, with a one-to-one correspondence between the discrete stencil coefficients and the differential operators of the governing equations. When combined with a displacement-function formulation that reduces the coupled equilibrium equations to a single scalar potential equation, the FDM provides a self-contained framework that is readily parameterised across material systems and boundary conditions. The present work develops and applies this framework to Al-based MMC plates under two distinct thermal loading configurations.

1.4 Outline Of Thesis

Chapter 2 reviews the relevant literature on orthotropic composite mechanics, thermal conduction, thermoelastic analysis, displacement function methods, and finite difference ap-

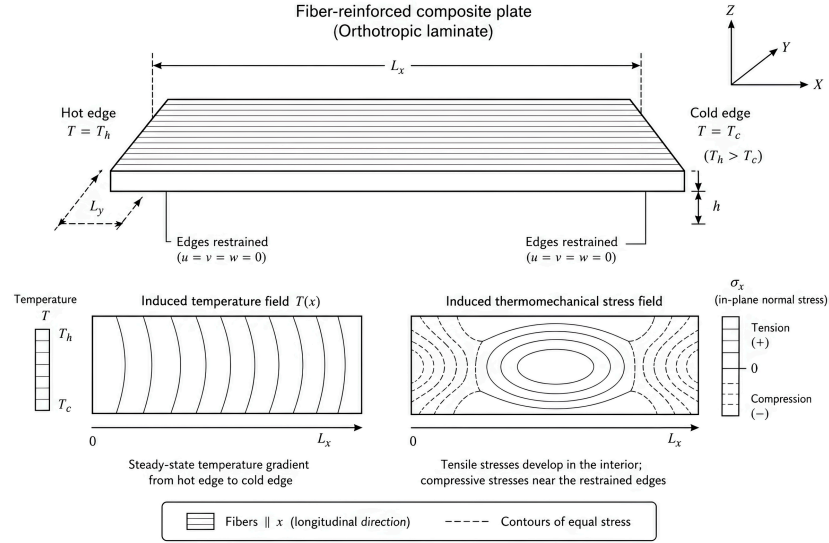


Figure 1.3: Rectangular orthotropic composite plate subjected to an edge temperature differential, illustrating the induced thermal gradient and the resulting constrained thermomechanical stress field.

proaches, and identifies the research gap addressed by the present work. The research objectives are also stated in Chapter 2. Chapter 3 presents the complete methodology: problem definition, governing equations, displacement-function formulations, finite difference discretisation, material properties, boundary conditions, and solver validation. Chapter 4 presents and discusses the thermomechanical stress results for both problems. Chapter 5 states the conclusions and recommendations for future work.

CHAPTER 2

LITERATURE REVIEW

2.1 Overview

Five areas of the literature bear directly on this work: fiber-reinforced composites in aerospace structures, thermal conduction in orthotropic media, thermoelastic analysis of composite panels, displacement function methods in plane elasticity, and finite difference discretization of coupled PDE systems. The identified research gap and the research objectives arising from it are stated at the end.

2.2 Orthotropic And Composite Materials In Aerospace Structures

Soutis (2005) traced the expansion of fiber-reinforced composites from control surfaces and fairings in early commercial aircraft to primary fuselage and wing structures in current designs, noting that the shift was driven by the specific stiffness advantage of carbon-fiber systems and their insensitivity to fatigue crack growth. Nurhaniza et al. (2010) used ABAQUS to analyse unidirectional E-glass laminates and found that tensile stiffness and thermal deformation are both strongly sensitive to fiber angle, with the 0° laminate considerably stiffer and less deformation-prone than 90° or 45° configurations.

Metal matrix composites extend this range to applications where polymer-matrix systems cannot sustain the required temperatures. Pooja et al. (2025) reviewed the fabrication and properties of aluminum-, copper-, magnesium-, titanium-, and zinc-based MMCs, finding that lower thermal expansion and improved wear resistance make aluminium-based systems particularly suited to components subjected to repeated thermal cycling. Their elastic and thermal properties are estimated through rule-of-mixtures relations (Gibson, 2016; Kaw, 2005).

2.3 Thermal Conduction In Orthotropic Media

The governing equation for steady-state heat conduction in an orthotropic medium involves a full second-rank conductivity tensor whose off-diagonal components are non-zero when the principal material axes are not aligned with the global coordinate axes.

Sameti et al. (2014) presented both analytical and finite difference solutions for steady-state heat conduction in an orthotropic rectangular domain, demonstrating that the off-diagonal conductivity component K^{xy} introduces oblique heat flow that cannot be captured by isotropic or axis-aligned orthotropic models. Their analytical approach relied on a coordinate rotation that transforms the orthotropic equation into an equivalent isotropic form, and their FDM results agreed with the analytical temperatures to within the expected second-order truncation error.

Prasanthi et al. (2025) measured longitudinal and transverse conductivities of bagasse fiber-reinforced epoxy composites at varying fiber weight fractions. Longitudinal conductivity rose with fiber content while transverse conductivity changed little, in line with the series and parallel micromechanical models used here (Kaw, 2005).

Hu et al. (2025) studied the temperature field in a composite wing electric heating anti-icing structure using an extended layerwise finite element method, incorporating convective boundary conditions and delamination-induced thermal discontinuities. Although their analysis addresses a more complex geometry than the rectangular panel considered here, the governing thermal field equation and the convective boundary condition formulation are directly analogous to those adopted in the present work.

2.4 Thermoelastic Analysis Of Composite Panels

The classical treatment of thermoelastic stress in orthotropic bodies is presented in Noda et al. (2003), who derived the general constitutive relations coupling the stress tensor to both mechanical strain and the thermal strain increment $\alpha_{ij} \Delta T$. For plane-stress problems — the configuration adopted in this thesis — the out-of-plane stress components vanish identically, and the in-plane thermoelastic constitutive equations reduce to a 3×3 stiffness system in which the transformed reduced stiffness coefficients $\bar{Q}_{ij}$ multiply the mechanical strains and the transformed thermal expansion coefficients $\bar{\alpha}_x, \bar{\alpha}_y, \bar{\alpha}_{xy}$ multiply the temperature change (Jones, 1999; Kaw, 2005).

Tezvergil et al. (2003) measured the linear coefficients of thermal expansion of unidirectional fiber-reinforced composites as functions of fiber orientation, confirming that off-axis laminates exhibit apparent shear-extension coupling in their thermal expansion response — a consequence of the non-zero $\bar{\alpha}_{xy}$ term that arises for $\theta \neq 0^\circ$ and $\theta \neq 90^\circ$. Cho (1999) applied finite element analysis to thermal stress in functionally graded com-

posites, highlighting that steep thermal gradients at material interfaces produce stress concentrations, and that the spatial distributions of temperature and stress are both sensitive to the material property variation.

More recently, Yang et al. (2025) developed a polygonal finite element method for steady-state and transient thermal stress analysis, demonstrating second-order convergence on non-matching meshes and confirming that for diffusion-dominated elliptic problems of the type considered here, second-order spatial accuracy is achievable with both finite difference and finite element discretizations of comparable stencil width.

2.5 Displacement Function Methods In Elasticity

The Airy stress function approach, in which a scalar potential $\psi(x, y)$ is introduced such that the in-plane stresses are expressed as its second derivatives, reduces the plane-stress equilibrium and compatibility equations to a single biharmonic equation for isotropic materials (Timoshenko and Goodier, 1970). For orthotropic materials under plane stress, the analogous displacement function formulation yields a fourth-order partial differential equation whose coefficients depend on the transformed reduced stiffness components $\bar{Q}_{ij}$ (Reddy, 2004). The inclusion of a thermal loading term introduces an additional right-hand side involving first derivatives of the temperature field, coupling the mechanical and thermal governing equations (Noda et al., 2003).

Ebrahimijahan et al. (2022) solved plane elastostatic equations for orthotropic functionally graded materials using an integrated radial basis function-finite difference method, demonstrating that the mixed cross-derivative terms $\partial^2/\partial x \partial y$ arising through the $\bar{Q}_{16}$ and $\bar{Q}_{26}$ stiffness coefficients can be discretized accurately using central-difference approximations on regular Cartesian grids. Ahmed et al. (2005) gave a general mathematical formulation for the finite-difference solution of mixed-boundary-value problems in anisotropic materials, providing the systematic treatment of combined displacement and stress conditions that the present work also requires. Building on this, the formulation has been applied to a range of orthotropic composite geometries: the elastic field of a unidirectional orthotropic cantilever beam (Afsar et al., 2008), stiffened orthotropic panels under uniaxial tension (Nath et al., 2007), and short guided columns under eccentric loading, where the single-potential reduction is used to handle a rigidly fixed supporting edge with fibers aligned along the member axis (Nath and Ahmed, 2008a), and stiffened orthotropic struts under eccentric loading (Nath and Ahmed, 2009). An analytical application of the

single-potential approach to two-dimensional mixed-boundary elastic problems is given by Idris et al. (1996). Akanda et al. (2002) demonstrated that the same displacement-potential function combined with finite differences resolves the stress field in a geometrically irregular, mixed-boundary domain (gear teeth), indicating that the method is not restricted to simple rectangular panels. The displacement function formulation adopted in this thesis follows the derivation of Kaw (2005) and Jones (1999), extended to include the thermoelastic body-force term.

2.6 Finite Difference Methods For Coupled PDE Systems

The theoretical foundation for the finite difference method applied to boundary value problems governed by elliptic partial differential equations is provided by LeVeque (2007), who established the conditions for consistency, stability, and convergence of second-order accurate schemes and derived truncation error bounds for central, forward, and backward difference approximations. For the steady-state problems considered in this thesis, the governing equations are elliptic, and the Lax equivalence theorem guarantees that consistency and stability together imply convergence (LeVeque, 2007).

Ailawalia et al. (2025) applied both explicit and implicit finite difference schemes to a two-dimensional coupled hygrothermoelastic boundary value problem, reporting close agreement between the two approaches and confirming the suitability of finite difference discretization for coupled thermo-mechanical systems on rectangular domains. For the steady-state formulation of the present work, a direct iterative solver is employed, and the nine-point and twenty-one-point stencils required by the orthotropic conduction and displacement-function equations are derived following the approach of Sameti et al. (2014) and Ebrahimijahan et al. (2022).

Ahmed et al. (2005) presented a general mathematical formulation for the finite-difference solution of mixed-boundary-value problems in anisotropic materials, addressing the unified imposition of displacement-type and stress-type conditions at boundary nodes that the present solver also demands along its partially fixed edges. The same displacement-potential-plus-finite-difference strategy has been applied to orthotropic composite members: the elastic field of an orthotropic cantilever beam (Afsar et al., 2008) and of a short orthotropic column under distributed load (Nath and Ahmed, 2008b), and to a non-rectangular mixed-boundary domain in the gear-tooth stress analysis of Akanda et al. (2002). Where displacement-potential solutions of this kind have been cross-checked

against finite-element predictions — as in the stiffened-strut analysis of Nath and Ahmed (2009) the close agreement supports the use of a numerical-versus-FEM comparison as a validation strategy of the kind adopted in Chapter 7 of this thesis. Chaudhari and Sutar (2021) applied the reduced differential transform method to a thermoelastic problem in the hyperbolic heat conduction regime, providing analytical comparison data for the steady-state thermoelastic solutions in the Fourier limit adopted here. Fang et al. (2024) applied ABAQUS finite element analysis to thermal stress in a multi-layer refractory structure, providing a methodology reference for the ABAQUS cross-validation study conducted in Section 3.8.

2.7 Identified Research Gap

The components of the present problem, orthotropic conduction, thermoelastic constitutive relations, displacement function formulations, and finite difference discretization, are each individually well established. Orthotropic steady-state conduction has analytical and finite-difference solutions (Sameti et al., 2014); the thermoelastic constitutive relations for orthotropic bodies are classical (Noda et al., 2003); and the displacement-potential function formulation, solved by finite differences under mixed boundary conditions, has been developed in detail for orthotropic composite members, cantilever beams (Afsar et al., 2008), stiffened panels (Nath et al., 2007), short columns (Nath and Ahmed, 2008a,b), stiffened struts under eccentric loading (Nath and Ahmed, 2009), two-dimensional mixed-boundary domains (Idris et al., 1996), and irregular mixed-boundary geometries (Akanda et al., 2002) on the general anisotropic-FDM foundation of Ahmed et al. (2005).

What has not been reported is the unification of these strands into a single from-first-principles FDM treatment of the *coupled* steady-state thermomechanical problem for an orthotropic composite panel. The existing displacement-potential studies solve purely mechanical loading and carry no thermal forcing term; the orthotropic conduction studies (Sameti et al., 2014) solve the thermal sub-problem alone; and published coupled analyses rely on commercial finite-element software (Fang et al., 2024; Nurhaniza et al., 2010; Yang et al., 2025), in which the coupling structure is not transparent. This thesis fills that gap: it extends the displacement-potential function formulation to include a thermoelastic forcing term derived from an orthotropic conduction field, implements the coupled system in a second-order FDM solver, validates it against commercial finite element solutions for Problem I, and applies it parametrically across a left-edge temperature sweep for Problem II.

2.8 Research Objectives

Based on the gap identified above, the specific objectives of this research are:

- (i) To derive the steady-state heat conduction equation for an orthotropic composite lamina.
- (ii) To formulate the plane-stress thermoelastic equilibrium equations for an orthotropic composite lamina and reduce the coupled system to a single governing equation through a displacement-function approach.
- (iii) To implement finite difference solutions for both governing equations and solve the coupled thermomechanical problem numerically.
- (iv) To conduct a parametric study of the thermomechanical stress field of the composite plate under varying material and loading conditions.

CHAPTER 3

METHODOLOGY

3.1 Overview

The physical problem and plate geometry are defined first, followed by estimation of the effective thermo-mechanical properties using the Rule of Mixtures. The mathematical derivations for the thermal and thermomechanical governing equations and their displacement-function formulations are then developed. The finite difference discretisation and solution procedure follow, after which the material properties and boundary conditions for both problems are tabulated. The chapter concludes with a grid convergence study and cross-verification against commercial finite element solutions.

3.2 Problem Description

The present work investigates the steady-state thermomechanical response of a rectangular orthotropic composite plate subjected to edge thermal loading. The plate is modelled in the x - y plane with in-plane dimensions $L_x = 3$ m and $L_y = 1$ m, giving an aspect ratio $L_x/L_y = 3$, and uniform thickness $w = 0.01$ m. The fiber axis is aligned with the global x -axis ($\theta = 0^\circ$) throughout, and the fiber volume fraction is $V_f = 0.50$ except where it is varied explicitly in the parametric study.

The composite system used for the detailed analysis is Al-B₄C: a 6061-T6 aluminium matrix reinforced with boron carbide fibers. This system is selected as the primary case on account of its intermediate thermal expansion characteristics among the aluminium-based MMC systems considered. The effective thermo-mechanical properties at $V_f = 0.50$ are listed in Table 3.1.

The plate geometry and coordinate system are shown in Figure 3.1.

Two boundary value problems with different thermal gradient directions and mechanical edge conditions are solved. Their full specifications are given in Sections 3.7.2 and 3.7.3.

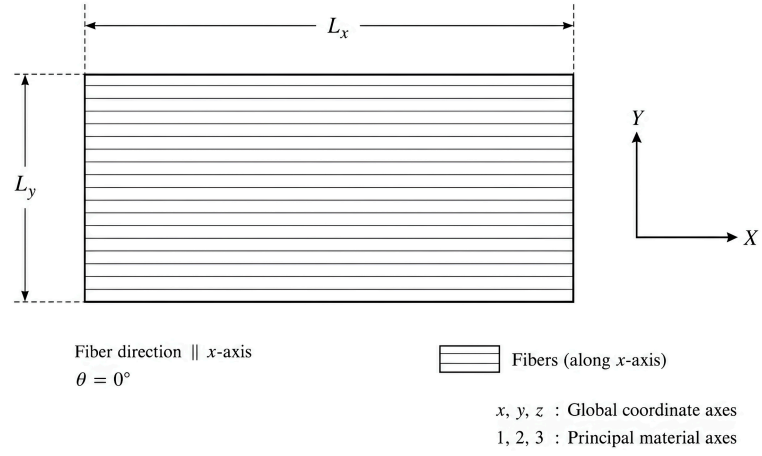


Figure 3.1: Rectangular composite plate domain with in-plane dimensions L_x and L_y ($L_x/L_y = 3$), showing the plate coordinate axes and the fiber direction aligned with the x -axis ($\theta = 0^\circ$).

3.3 Material Property Estimation

The effective thermo-mechanical properties of the composite lamina are estimated from the constituent fiber and matrix properties using the Rule of Mixtures. The elastic moduli, Poisson's ratios, coefficients of thermal expansion, and thermal conductivities are required as inputs to both the thermal and thermomechanical governing equations derived in Sections 3.4 and 3.5.

3.3.1 Effective Elastic Properties Via Rule Of Mixtures

For a void-free composite lamina, the fiber and matrix volume fractions satisfy the constraint $V_f + V_m = 1$. The longitudinal modulus and the major Poisson's ratio are governed by the parallel (Voigt) rule of mixtures (Kaw, 2005):

$$E_1 = V_f E_f + V_m E_m, \quad \nu_{12} = V_f \nu_f + V_m \nu_m \quad (3.1)$$

The transverse modulus and the in-plane shear modulus are governed by the series (Reuss) inverse rule of mixtures (Gibson, 2016):

$$E_2 = \left(\frac{V_f}{E_f} + \frac{V_m}{E_m} \right)^{-1}, \quad G_{12} = \left(\frac{V_f}{G_f} + \frac{V_m}{G_m} \right)^{-1} \quad (3.2)$$

The minor Poisson's ratio follows from symmetry: $\nu_{21} = \nu_{12} E_2/E_1$. Numerical values for all four material systems are given in Section 3.7.

3.3.2 Coefficients Of Thermal Expansion

The principal coefficients of thermal expansion (CTEs) of the composite lamina in the fiber direction and the transverse direction are estimated from the constituent properties by (Gibson, 2016):

$$\alpha_1 = \frac{V_f E_f \alpha_f + V_m E_m \alpha_m}{E_1}, \quad \alpha_2 = (1 + \nu_f) V_f \alpha_f + (1 + \nu_m) V_m \alpha_m - \nu_{12} \alpha_1 \quad (3.3)$$

Because the fiber axis coincides with the x -axis, α_1 and α_2 enter the constitutive relations (3.11) directly. Numerical values appear in Section 3.7.

3.3.3 Effective Thermal Conductivities

The thermal conductivities of the composite in the fiber direction and the transverse direction are estimated by the parallel and series rules of mixtures, respectively (Kaw, 2005):

$$k_1 = V_f k_f + V_m k_m, \quad k_2 = \left(\frac{V_f}{k_f} + \frac{V_m}{k_m} \right)^{-1} \quad (3.4)$$

With axis-aligned fibers, k_1 governs conduction in the x -direction and k_2 in the y -direction; no off-diagonal term appears. Numerical values are given in Section 3.7.

Figure 3.2 shows the variation of the effective properties with fiber volume fraction.

3.4 Mathematical Modelling Of The Thermal Problem

The steady-state temperature distribution in the plate is governed by an orthotropic heat conduction equation derived from an energy balance over a differential control volume. The thermal boundary condition types and the modelling assumptions on which the governing equation rests are stated subsequently.

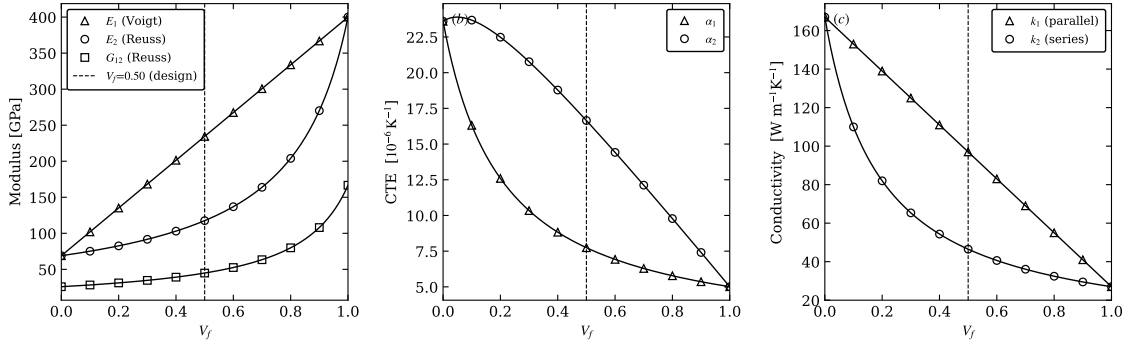


Figure 3.2: Effective thermo-mechanical properties of Al-B₄C as functions of fiber volume fraction V_f , computed using the Rule of Mixtures. (a) Elastic moduli E_1 (Voigt), E_2 and G_{12} (Reuss); (b) coefficients of thermal expansion α_1 and α_2 ; (c) thermal conductivities k_1 (parallel) and k_2 (series). Filled markers indicate the design value $V_f = 0.50$ tabulated in Table 3.1.

3.4.1 Steady-State Governing Equation

For an orthotropic lamina with fiber axis aligned with the x -coordinate, the Fourier heat flux components are (Noda et al., 2003):

$$q_x = -k_1 \frac{\partial T}{\partial x} \quad (3.5a)$$

$$q_y = -k_2 \frac{\partial T}{\partial y} \quad (3.5b)$$

Applying an energy balance over a control volume $\Delta x \Delta y w$ and accounting for convective losses from the top and bottom faces (see Appendix A.6 for the step-by-step derivation) yields the following steady-state heat conduction–convection equation (Noda et al., 2003):

$$\boxed{k_1 \frac{\partial^2 T}{\partial x^2} + k_2 \frac{\partial^2 T}{\partial y^2} - \frac{2h_c}{w}(T - T_0) + g = 0} \quad (3.6)$$

Here g is the volumetric rate of internal heat generation (W m^{-3}), h_c is the surface convection coefficient ($\text{W m}^{-2} \text{K}^{-1}$), w is the panel thickness, and T_0 is the ambient reference temperature. Equation (3.6) is an elliptic partial differential equation that is discretized by a five-point finite difference stencil in Section 3.6.

3.4.2 Thermal Boundary Conditions

Thermal boundary conditions at the panel edges are required to obtain a unique solution of (3.6). These conditions specify the thermal interaction between the panel boundary and its environment and fall into three standard categories (Noda et al., 2003):

- (i) *Type I — Isothermal (Dirichlet)*. The temperature is prescribed at the boundary:

$$T = T_0 \quad (3.7)$$

where T_0 is the specified boundary temperature. This condition applies when an edge is in contact with a thermal reservoir of known temperature.

- (ii) *Type II — Prescribed Heat Flux (Neumann)*. The normal component of the heat flux at the boundary is specified:

$$-k_n \frac{\partial T}{\partial n} = q_n \quad (3.8)$$

where n denotes the outward normal to the boundary, k_n is the conductivity in the normal direction, and q_n is the prescribed heat flux (positive for inflow, negative for outflow). The special case $q_n = 0$ gives an adiabatic boundary.

- (iii) *Type III — Convective (Robin / Mixed)*. The heat flux at the boundary is proportional to the difference between the surface temperature and the ambient temperature:

$$-k_n \frac{\partial T}{\partial n} = h_c (T - T_0) \quad (3.9)$$

where h_c is the convective heat transfer coefficient at the edge and T_0 is the ambient temperature. This condition models convective heat exchange at a free surface.

The specific combination of boundary conditions applied to each panel edge in the numerical experiments is described in Section 3.7.

3.4.3 Modelling Assumptions

The governing equation (3.6) rests on the following assumptions:

- (i) *Steady state.* All time derivatives are zero; the analysis yields the equilibrium temperature distribution after thermal transients have decayed.
- (ii) *Axis-aligned fibers.* The fiber axis coincides with the global x -axis, so the conductivity tensor is diagonal: k_1 acts in x and k_2 acts in y . No cross-derivative conduction term appears.
- (iii) *Uniform in-plane conductivities.* k_1 and k_2 are treated as spatially constant, computed from the rule-of-mixtures expressions in Section 3.3.3.
- (iv) *Lumped through-thickness convection.* The panel is thin and the temperature field is treated as two-dimensional. Surface convection from the top and bottom faces is captured by the lumped term $2h_c/w$ in (3.6).
- (v) *Linear surface convection.* Heat loss at the top and bottom surfaces follows Newton's law of cooling, valid when h_c is independent of temperature.
- (vi) *Negligible radiation.* Radiative heat transfer at the surfaces is assumed negligible relative to convection and conduction.
- (vii) *Uniform internal heat generation.* The volumetric source g is spatially constant; the adiabatic case $g = 0$ is the default unless stated otherwise.

3.5 Mathematical Modelling Of The Thermomechanical Problem

The stress field induced by the temperature distribution obtained from Section 3.4 is determined under the assumption of plane stress. The in-plane stress state of the composite lamina is characterised by the components σ_{xx} , σ_{yy} , and τ_{xy} , which are governed by the coupled thermoelastic equilibrium and constitutive equations derived in this section. A scalar displacement potential function $\psi(x, y)$ is introduced to reduce the coupled system to a single fourth-order governing equation, and two independent formulations are obtained for thermal gradients applied in each coordinate direction.

In a Cartesian coordinate system, equilibrium of the stress field in the absence of body forces requires (Timoshenko and Goodier, 1970):

$$\frac{\partial \sigma_{xx}}{\partial x} + \frac{\partial \tau_{xy}}{\partial y} = 0 \quad (3.10a)$$

$$\frac{\partial \tau_{xy}}{\partial x} + \frac{\partial \sigma_{yy}}{\partial y} = 0 \quad (3.10b)$$

The orthotropic characteristics of the material are described using the reduced stiffness matrix $[Q]$, which relates the stress components to the strains and to the temperature change ΔT . This is expressed in Equation (3.11), encompassing both elastic and thermal effects (Kaw, 2005):

$$\begin{Bmatrix} \sigma_{xx} \\ \sigma_{yy} \\ \tau_{xy} \end{Bmatrix} = \begin{bmatrix} Q_{11} & Q_{12} & 0 \\ Q_{12} & Q_{22} & 0 \\ 0 & 0 & Q_{66} \end{bmatrix} \begin{Bmatrix} \varepsilon_x - \alpha_1 \Delta T \\ \varepsilon_y - \alpha_2 \Delta T \\ \gamma_{xy} \end{Bmatrix} \quad (3.11)$$

where α_1 and α_2 are the coefficients of thermal expansion (CTE) in the principal fiber and transverse directions, respectively. The stiffness parameters Q_{11} , Q_{22} , Q_{12} , and Q_{66} are related to the elastic moduli E_1 , E_2 , Poisson's ratios ν_{12} , ν_{21} , and the shear modulus G_{12} (Jones, 1999):

$$Q_{11} = \frac{E_1}{1 - \nu_{12}\nu_{21}}, \quad Q_{22} = \frac{E_2}{1 - \nu_{12}\nu_{21}}, \quad Q_{12} = \frac{\nu_{12}E_2}{1 - \nu_{12}\nu_{21}}, \quad Q_{66} = G_{12} \quad (3.12)$$

Using the linearised strain-displacement relations (Jones, 1999):

$$\varepsilon_x = \frac{\partial u_x}{\partial x}, \quad \varepsilon_y = \frac{\partial u_y}{\partial y}, \quad \gamma_{xy} = \frac{\partial u_x}{\partial y} + \frac{\partial u_y}{\partial x} \quad (3.13)$$

the stress components expressed in terms of the displacement components are (Kaw, 2005):

$$\sigma_{xx} = Q_{11} \frac{\partial u_x}{\partial x} + Q_{12} \frac{\partial u_y}{\partial y} - (Q_{11}\alpha_1 + Q_{12}\alpha_2) \Delta T \quad (3.14a)$$

$$\sigma_{yy} = Q_{12} \frac{\partial u_x}{\partial x} + Q_{22} \frac{\partial u_y}{\partial y} - (Q_{12}\alpha_1 + Q_{22}\alpha_2) \Delta T \quad (3.14b)$$

$$\tau_{xy} = Q_{66} \left(\frac{\partial u_x}{\partial y} + \frac{\partial u_y}{\partial x} \right) \quad (3.14c)$$

Substituting (3.14) into the equilibrium conditions (3.10) yields two coupled second-order partial differential equations in the displacement components (Kaw, 2005):

$$Q_{11} \frac{\partial^2 u_x}{\partial x^2} + (Q_{12} + Q_{66}) \frac{\partial^2 u_y}{\partial x \partial y} + Q_{66} \frac{\partial^2 u_x}{\partial y^2} = (Q_{11}\alpha_1 + Q_{12}\alpha_2) \frac{\partial T}{\partial x} \quad (3.15a)$$

$$Q_{66} \frac{\partial^2 u_y}{\partial x^2} + (Q_{12} + Q_{66}) \frac{\partial^2 u_x}{\partial x \partial y} + Q_{22} \frac{\partial^2 u_y}{\partial y^2} = (Q_{12}\alpha_1 + Q_{22}\alpha_2) \frac{\partial T}{\partial y} \quad (3.15b)$$

To simplify the analysis and reduce the coupled system to a single equation, a scalar displacement-potential function $\psi(x, y)$ is introduced. This follows an established displacement potential approach to two-dimensional orthotropic elasticity, in which the in-plane displacement components are expressed through derivatives of a single function so that two of the equilibrium equations are satisfied identically while the third becomes the governing equation for ψ (Afsar et al., 2008; Idris et al., 1996; Nath and Ahmed, 2008a). The displacement components are written in terms of the second derivatives of ψ with constants β_1 to β_6 to be fixed by the problem configuration (Kaw, 2005). Substituting this ansatz into the equilibrium equations yields two fourth-order equations in ψ . The present formulation extends this purely mechanical construction by retaining the thermal forcing terms on the right-hand sides, so that the temperature gradient enters the displacement-potential equation directly — the feature that distinguishes the coupled thermoelastic problem of this thesis from the mechanical-load displacement-potential solutions in the literature (Ahmed et al., 2005; Akanda et al., 2002).

$$u_x = \beta_1 \frac{\partial^2 \psi}{\partial x^2} + \beta_2 \frac{\partial^2 \psi}{\partial x \partial y} + \beta_3 \frac{\partial^2 \psi}{\partial y^2} \quad (3.16a)$$

$$u_y = \beta_4 \frac{\partial^2 \psi}{\partial x^2} + \beta_5 \frac{\partial^2 \psi}{\partial x \partial y} + \beta_6 \frac{\partial^2 \psi}{\partial y^2} \quad (3.16b)$$

Substituting (3.16) into (3.15) yields the two equilibrium equations expressed in terms of the controlling function $\psi(x, y)$:

$$\begin{aligned}
& \beta_1 Q_{11} \frac{\partial^4 \psi}{\partial x^4} + \{\beta_2 Q_{11} + \beta_4 (Q_{12} + Q_{66})\} \frac{\partial^4 \psi}{\partial x^3 \partial y} \\
& + \{\beta_1 Q_{66} + \beta_3 Q_{11} + \beta_5 (Q_{12} + Q_{66})\} \frac{\partial^4 \psi}{\partial x^2 \partial y^2} \\
& + \{\beta_2 Q_{66} + \beta_6 (Q_{12} + Q_{66})\} \frac{\partial^4 \psi}{\partial x \partial y^3} + \beta_3 Q_{66} \frac{\partial^4 \psi}{\partial y^4} \\
& = (Q_{11} \alpha_1 + Q_{12} \alpha_2) \frac{\partial T}{\partial x}
\end{aligned} \tag{3.17a}$$

$$\begin{aligned}
& \beta_4 Q_{66} \frac{\partial^4 \psi}{\partial x^4} + \{\beta_1 (Q_{12} + Q_{66}) + \beta_5 Q_{66}\} \frac{\partial^4 \psi}{\partial x^3 \partial y} \\
& + \{\beta_2 (Q_{12} + Q_{66}) + \beta_4 Q_{22} + \beta_6 Q_{66}\} \frac{\partial^4 \psi}{\partial x^2 \partial y^2} \\
& + \{\beta_3 (Q_{12} + Q_{66}) + \beta_5 Q_{22}\} \frac{\partial^4 \psi}{\partial x \partial y^3} + \beta_6 Q_{22} \frac{\partial^4 \psi}{\partial y^4} \\
& = (Q_{12} \alpha_1 + Q_{22} \alpha_2) \frac{\partial T}{\partial y}
\end{aligned} \tag{3.17b}$$

These two fourth-order partial differential equations are the equilibrium conditions for the coupled thermoelastic problem expressed through the displacement function ψ . The temperature gradients in the two directions are addressed individually using two distinct formulations based on the same displacement function.

3.5.1 Formulation I

Formulation I pertains to the scenario in which Equation (3.17a) is regarded as the governing differential equation for ψ , incorporating the temperature gradient in the x -direction. For (3.17a) to serve as the governing equation, Equation (3.17b) must be universally satisfied. This occurs when all coefficients on the left-hand side of (3.17b) are independently zero and the temperature gradient in the y -direction is absent; the mathematical conditions

for this are (Kaw, 2005):

$$\beta_4 = 0, \quad \beta_6 = 0 \quad (3.18a)$$

$$\beta_1(Q_{12} + Q_{66}) + \beta_5 Q_{66} = 0 \quad (3.18b)$$

$$\beta_2(Q_{12} + Q_{66}) + \beta_6 Q_{66} = 0 \quad (3.18c)$$

$$\beta_3(Q_{12} + Q_{66}) + \beta_5 Q_{22} = 0 \quad (3.18d)$$

Setting $\beta_5 = 1$ and solving (3.18):

$$\beta_1 = -\frac{Q_{66}}{Q_{12} + Q_{66}}, \quad \beta_2 = \beta_4 = \beta_6 = 0, \quad \beta_3 = -\frac{Q_{22}}{Q_{12} + Q_{66}} \quad (3.19)$$

Substituting (3.19) into (3.17a) gives the governing equation for Formulation I (Kaw, 2005):

$$\begin{aligned} \frac{\partial^4 \psi}{\partial x^4} + \left(\frac{Q_{11}Q_{22} - Q_{12}^2 - 2Q_{12}Q_{66}}{Q_{11}Q_{66}} \right) \frac{\partial^4 \psi}{\partial x^2 \partial y^2} + \frac{Q_{22}}{Q_{11}} \frac{\partial^4 \psi}{\partial y^4} \\ = -\frac{(Q_{12} + Q_{66})(Q_{11}\alpha_1 + Q_{12}\alpha_2)}{Q_{11}Q_{66}} \frac{\partial T}{\partial x} \end{aligned} \quad (3.20)$$

The thermoelastic problem is thereby reduced to determining a single scalar function ψ from Equation (3.20), with the temperature gradient in the y -direction absent. Substituting (3.19) into (3.16) gives the displacement components:

$$u_x = -\frac{Q_{66}}{Q_{12} + Q_{66}} \frac{\partial^2 \psi}{\partial x^2} - \frac{Q_{22}}{Q_{12} + Q_{66}} \frac{\partial^2 \psi}{\partial y^2} \quad (3.21a)$$

$$u_y = \frac{\partial^2 \psi}{\partial x \partial y} \quad (3.21b)$$

Substituting (3.21) into (3.14):

$$\begin{aligned}\sigma_{xx} = & -\frac{Q_{66}}{Q_{12} + Q_{66}} \frac{\partial^3 \psi}{\partial x^3} \\ & + \frac{Q_{12}^2 + Q_{12}Q_{66} - Q_{11}Q_{22}}{Q_{12} + Q_{66}} \frac{\partial^3 \psi}{\partial x \partial y^2} \\ & - (Q_{11}\alpha_1 + Q_{12}\alpha_2) \Delta T\end{aligned}\quad (3.22a)$$

$$\sigma_{yy} = -\frac{Q_{12}Q_{66}}{Q_{12} + Q_{66}} \frac{\partial^3 \psi}{\partial x^3} + \frac{Q_{22}Q_{66}}{Q_{12} + Q_{66}} \frac{\partial^3 \psi}{\partial x \partial y^2} - (Q_{12}\alpha_1 + Q_{22}\alpha_2) \Delta T \quad (3.22b)$$

$$\tau_{xy} = \frac{Q_{12}Q_{66}}{Q_{12} + Q_{66}} \frac{\partial^3 \psi}{\partial x^2 \partial y} - \frac{Q_{22}Q_{66}}{Q_{12} + Q_{66}} \frac{\partial^3 \psi}{\partial y^3} \quad (3.22c)$$

The strain components follow from (3.13) and (3.21):

$$\varepsilon_x = -\frac{Q_{66}}{Q_{12} + Q_{66}} \frac{\partial^3 \psi}{\partial x^3} - \frac{Q_{22}}{Q_{12} + Q_{66}} \frac{\partial^3 \psi}{\partial x \partial y^2} \quad (3.23a)$$

$$\varepsilon_y = \frac{\partial^3 \psi}{\partial x \partial y^2} \quad (3.23b)$$

$$\gamma_{xy} = \frac{Q_{12}}{Q_{12} + Q_{66}} \frac{\partial^3 \psi}{\partial x^2 \partial y} - \frac{Q_{22}}{Q_{12} + Q_{66}} \frac{\partial^3 \psi}{\partial y^3} \quad (3.23c)$$

The β_i coefficients for Formulation I are obtained by substituting (3.16) into the equilibrium equations and setting $\beta_5 = 1$:

$$\beta_4 = 0, \quad \beta_1 = -\frac{Q_{66}}{Q_{12} + Q_{66}}, \quad \beta_2 = \beta_6 = 0, \quad \beta_3 = -\frac{Q_{22}}{Q_{12} + Q_{66}} \quad (3.24)$$

The coefficients C_1 and C_2 appearing in the governing equation (3.20) are confirmed as:

$$C_1 = \frac{Q_{11}Q_{22} - Q_{12}^2 - 2Q_{12}Q_{66}}{Q_{11}Q_{66}}, \quad C_2 = \frac{Q_{22}}{Q_{11}} \quad (3.25)$$

3.5.2 Formulation II

Formulation II considers Equation (3.17b) as the governing equation for $\psi(x, y)$, incorporating the temperature gradient in the y -direction. The first equilibrium equation (3.17a)

must be universally satisfied; the conditions for this are (Kaw, 2005):

$$\beta_1 = 0, \quad \beta_3 = 0 \quad (3.26a)$$

$$\beta_2 Q_{11} + \beta_4(Q_{12} + Q_{66}) = 0 \quad (3.26b)$$

$$\beta_1 Q_{66} + \beta_3 Q_{11} + \beta_5(Q_{12} + Q_{66}) = 0 \quad (3.26c)$$

$$\beta_2 Q_{66} + \beta_6(Q_{12} + Q_{66}) = 0 \quad (3.26d)$$

Setting $\beta_2 = 1$ and solving (3.26):

$$\beta_1 = \beta_3 = \beta_5 = 0, \quad \beta_4 = -\frac{Q_{11}}{Q_{12} + Q_{66}}, \quad \beta_6 = -\frac{Q_{66}}{Q_{12} + Q_{66}} \quad (3.27)$$

Substituting (3.27) into (3.17b) gives the governing equation for Formulation II, applicable in the absence of a temperature gradient in the x -direction (Kaw, 2005):

$$\begin{aligned} \frac{\partial^4 \psi}{\partial x^4} + \left(\frac{Q_{11}Q_{22} - Q_{12}^2 - 2Q_{12}Q_{66}}{Q_{11}Q_{66}} \right) \frac{\partial^4 \psi}{\partial x^2 \partial y^2} + \frac{Q_{22}}{Q_{11}} \frac{\partial^4 \psi}{\partial y^4} \\ = -\frac{(Q_{12} + Q_{66})(Q_{12}\alpha_1 + Q_{22}\alpha_2)}{Q_{11}Q_{66}} \frac{\partial T}{\partial y} \end{aligned} \quad (3.28)$$

The displacement components are:

$$u_x = \frac{\partial^2 \psi}{\partial x \partial y} \quad (3.29a)$$

$$u_y = -\frac{Q_{11}}{Q_{12} + Q_{66}} \frac{\partial^2 \psi}{\partial x^2} - \frac{Q_{66}}{Q_{12} + Q_{66}} \frac{\partial^2 \psi}{\partial y^2} \quad (3.29b)$$

Substituting (3.29) into (3.14):

$$\sigma_{xx} = \frac{Q_{11}Q_{66}}{Q_{12} + Q_{66}} \frac{\partial^3 \psi}{\partial x^2 \partial y} - \frac{Q_{12}Q_{66}}{Q_{12} + Q_{66}} \frac{\partial^3 \psi}{\partial y^3} - (Q_{11}\alpha_1 + Q_{12}\alpha_2) \Delta T \quad (3.30a)$$

$$\begin{aligned} \sigma_{yy} = \frac{Q_{12}^2 + Q_{12}Q_{66} - Q_{11}Q_{22}}{Q_{12} + Q_{66}} \frac{\partial^3 \psi}{\partial x^2 \partial y} \\ - \frac{Q_{22}Q_{66}}{Q_{12} + Q_{66}} \frac{\partial^3 \psi}{\partial y^3} - (Q_{12}\alpha_1 + Q_{22}\alpha_2) \Delta T \end{aligned} \quad (3.30b)$$

$$\tau_{xy} = \frac{Q_{12}Q_{66}}{Q_{12} + Q_{66}} \frac{\partial^3 \psi}{\partial x \partial y^2} - \frac{Q_{11}Q_{66}}{Q_{12} + Q_{66}} \frac{\partial^3 \psi}{\partial x^3} \quad (3.30c)$$

The strain components are:

$$\varepsilon_x = \frac{\partial^3 \psi}{\partial x^2 \partial y} \quad (3.31a)$$

$$\varepsilon_y = -\frac{Q_{11}}{Q_{12} + Q_{66}} \frac{\partial^3 \psi}{\partial x^2 \partial y} - \frac{Q_{66}}{Q_{12} + Q_{66}} \frac{\partial^3 \psi}{\partial y^3} \quad (3.31b)$$

$$\gamma_{xy} = -\frac{Q_{11}}{Q_{12} + Q_{66}} \frac{\partial^3 \psi}{\partial x^3} + \frac{Q_{12}}{Q_{12} + Q_{66}} \frac{\partial^3 \psi}{\partial x \partial y^2} \quad (3.31c)$$

The β_i coefficients for Formulation II are obtained by substituting (3.16) into (3.17a) and setting $\beta_2 = 1$:

$$\beta_1 = \beta_3 = \beta_5 = 0, \quad \beta_4 = -\frac{Q_{11}}{Q_{12} + Q_{66}}, \quad \beta_6 = -\frac{Q_{66}}{Q_{12} + Q_{66}} \quad (3.32)$$

Substituting (3.32) into (3.17b) recovers (3.28) with the same C_1 and C_2 as (3.25), confirming internal consistency between the two formulations.

The right-hand sides of the governing equations (3.20) and (3.28) are obtained by substituting the respective β_i values into the thermal terms. For Formulation I:

$$\text{RHS}_I = -\frac{Q_{12} + Q_{66}}{Q_{11}Q_{66}}(Q_{11}\alpha_1 + Q_{12}\alpha_2) \frac{\partial T}{\partial x} \quad (3.33)$$

For Formulation II:

$$\text{RHS}_{II} = -\frac{Q_{12} + Q_{66}}{Q_{11}Q_{66}}(Q_{12}\alpha_1 + Q_{22}\alpha_2) \frac{\partial T}{\partial y} \quad (3.34)$$

This chapter describes the finite difference discretization of the two governing equations derived in Chapter 3, and the assembly and solution of the resulting linear systems. The thermal governing equation (3.6) and the displacement-function governing equations (3.20) and (3.28) are each elliptic partial differential equations defined on a rectangular domain $[0, L_x] \times [0, L_y]$ with mixed boundary conditions. A uniform rectangular grid of $N_x \times N_y$ nodes with constant spacings $h = L_x/(N_x - 1)$ and $k = L_y/(N_y - 1)$ is used for both sub-problems. Interior nodes are discretized using central-difference approximations of second-order accuracy $\mathcal{O}(h^2)$; boundary and near-boundary nodes use forward or backward differences of the same order.

3.6 Numerical Solution

The governing equations derived in Sections 3.4 and 3.5 are solved numerically using the finite difference method on a uniform rectangular grid. The discretisation of both equations is described first, followed by the treatment of boundary conditions at the grid boundary nodes, and the procedure by which the resulting linear systems are assembled and solved.

3.6.1 Grid Layout And Node Numbering

The domain is discretized into a uniform grid with nodes at positions $(x_i, y_j) = ((i - 1)h, (j - 1)k)$ for $i = 1, 2, \dots, N_x$ and $j = 1, 2, \dots, N_y$. The two-dimensional node array is mapped to a one-dimensional index by $n = (j - 1)N_x + i$, giving a total of $N = N_x \times N_y$ unknowns per field. The boundary nodes occupy the four edge rows and columns: $i = 1$ (left), $i = N_x$ (right), $j = 1$ (bottom), $j = N_y$ (top). The unknown vectors for the thermal field T_{ij} and the displacement potential ψ_{ij} are each assembled into column vectors of length N , and the discretized equations are written as sparse linear systems $[\mathbf{A}]\{T\} = \{b\}$ and $[\mathbf{K}]\{\psi\} = \{f\}$ respectively.

3.6.2 Discretisation Of The Thermal Equation

The governing equation (3.6) is discretised on the uniform rectangular grid of Section 3.6.1 using second-order central differences. Applying central differences to the second-order derivatives in the thermal governing equation (3.6) at an interior node (i, j) gives (LeVeque, 2007):

$$\frac{k_1}{h^2}(T_{i+1,j} - 2T_{i,j} + T_{i-1,j}) + \frac{k_2}{k^2}(T_{i,j+1} - 2T_{i,j} + T_{i,j-1}) - \frac{2h_c}{w}T_{i,j} = -g - \frac{2h_c}{w}T_0 \quad (3.35)$$

Equation (3.35) is a five-point stencil involving the node itself and its four immediate neighbours. Collecting terms for $T_{i,j}$ and its four neighbours gives the standard form:

$$\begin{aligned} \frac{k_1}{h^2}T_{i-1,j} + \frac{k_2}{k^2}T_{i,j-1} - 2\left(\frac{k_1}{h^2} + \frac{k_2}{k^2} + \frac{h_c}{w}\right)T_{i,j} \\ + \frac{k_2}{k^2}T_{i,j+1} + \frac{k_1}{h^2}T_{i+1,j} = -g - \frac{2h_c}{w}T_0 \end{aligned} \quad (3.36)$$

Boundary nodes at which the temperature is prescribed (Type I, Dirichlet) are handled by replacing the corresponding equation with the identity $T_{i,j} = T_0$, which sets the diagonal entry to unity and the right-hand side to the prescribed value, leaving all other coefficients in that row zero. For boundary segments where convective exchange applies (Type III, Robin), the normal derivative is approximated by a one-sided first-order difference and the resulting equation is added to the stencil at the boundary node.

3.6.3 Discretisation Of The Displacement-Function Equation

Both Formulation I (3.20) and Formulation II (3.28) have the same left-hand operator:

$$\mathcal{L}[\psi] = \frac{\partial^4 \psi}{\partial x^4} + C_1 \frac{\partial^4 \psi}{\partial x^2 \partial y^2} + C_2 \frac{\partial^4 \psi}{\partial y^4} \quad (3.37)$$

The three fourth-order derivatives are approximated at interior node (i, j) by central differences of order $\mathcal{O}(h^2)$ (LeVeque, 2007):

$$\left. \frac{\partial^4 \psi}{\partial x^4} \right|_{i,j} \approx \frac{\psi_{i-2,j} - 4\psi_{i-1,j} + 6\psi_{i,j} - 4\psi_{i+1,j} + \psi_{i+2,j}}{h^4} \quad (3.38a)$$

$$\left. \frac{\partial^4 \psi}{\partial y^4} \right|_{i,j} \approx \frac{\psi_{i,j-2} - 4\psi_{i,j-1} + 6\psi_{i,j} - 4\psi_{i,j+1} + \psi_{i,j+2}}{k^4} \quad (3.38b)$$

$$\begin{aligned} \left. \frac{\partial^4 \psi}{\partial x^2 \partial y^2} \right|_{i,j} \approx & \frac{1}{h^2 k^2} \left(\psi_{i+1,j+1} - 2\psi_{i,j+1} + \psi_{i-1,j+1} \right. \\ & - 2(\psi_{i+1,j} - 2\psi_{i,j} + \psi_{i-1,j}) \\ & \left. + \psi_{i+1,j-1} - 2\psi_{i,j-1} + \psi_{i-1,j-1} \right) \end{aligned} \quad (3.38c)$$

The full stencil for $\mathcal{L}[\psi]$ therefore involves thirteen nodes: the five nodes in the standard cross plus the eight diagonal and second-neighbour nodes that appear in (3.38a)–(3.38c). At interior nodes two grid spacings from a boundary, one or more of the stencil nodes falls outside the domain; these are replaced by forward or backward one-sided differences of matching accuracy (LeVeque, 2007).

The right-hand side of the governing equation requires the first derivative of the temperature field. For Formulation I (3.20), the forcing is proportional to $\partial T / \partial x$, approximated

at interior node (i, j) by:

$$\left. \frac{\partial T}{\partial x} \right|_{i,j} \approx \frac{T_{i+1,j} - T_{i-1,j}}{2h} \quad (3.39)$$

For Formulation II (3.28), the forcing is proportional to $\partial T / \partial y$:

$$\left. \frac{\partial T}{\partial y} \right|_{i,j} \approx \frac{T_{i,j+1} - T_{i,j-1}}{2k} \quad (3.40)$$

The temperature field T_{ij} required here is the numerical solution of the thermal sub-problem from Section 3.6.2. The sequential solution strategy — thermal field first, then mechanical field — is valid because the problem is one-way coupled: temperature drives stress, but (under the linear elasticity and small-deformation assumptions of this study) the stress state does not feed back to alter the temperature distribution (Noda et al., 2003).

3.6.4 Boundary Conditions For The Displacement-Function Problem

The mechanical boundary conditions described in Section 3.7 specify either fixed displacements ($u_x = u_y = 0$) or zero stress ($\sigma_{nn} = \tau_{nt} = 0$) on each boundary segment. These physical conditions must be translated into conditions on the displacement function ψ and its derivatives.

For the *fixed* segments (inner 60% of each edge), the zero-displacement conditions $u_x = u_y = 0$ are imposed directly from the displacement expressions (3.21) or (3.29) by setting the relevant second derivatives of ψ to zero at each boundary node, reducing to Dirichlet conditions on the normal and tangential derivatives of ψ .

For the *stress-free* segments (outer 20% of each edge), the conditions $\sigma_{nn} = 0$ and $\tau_{nt} = 0$ are imposed by substituting the stress expressions (3.22) or (3.30) and setting the relevant combinations of third derivatives of ψ to zero, which are discretized with one-sided differences at the boundary nodes. The complete boundary stencils are derived from the stress expressions (3.22) and (3.30).

3.6.5 Assembly And Solution

For each field (thermal and mechanical), the discretized equations at all $N = N_x \times N_y$ nodes are assembled row by row into a sparse matrix $\mathbf{A}$ (or $\mathbf{K}$) and a right-hand side vector

b (or **f**). Both matrices are banded and diagonally dominant for the interior equations; the bandwidth depends on the stencil reach and the node-ordering scheme.

The linear systems are solved using a sparse direct solver, which selects the appropriate factorisation method based on the matrix structure. The thermal system is solved first; the resulting nodal temperature vector is then used to construct the right-hand side of the mechanical system, which is solved in turn. Stress and strain components at every node are recovered by applying the differentiation formulas (3.22) or (3.30) using central differences on the converged ψ field.

The complete solution procedure is summarised in Figure 3.3.

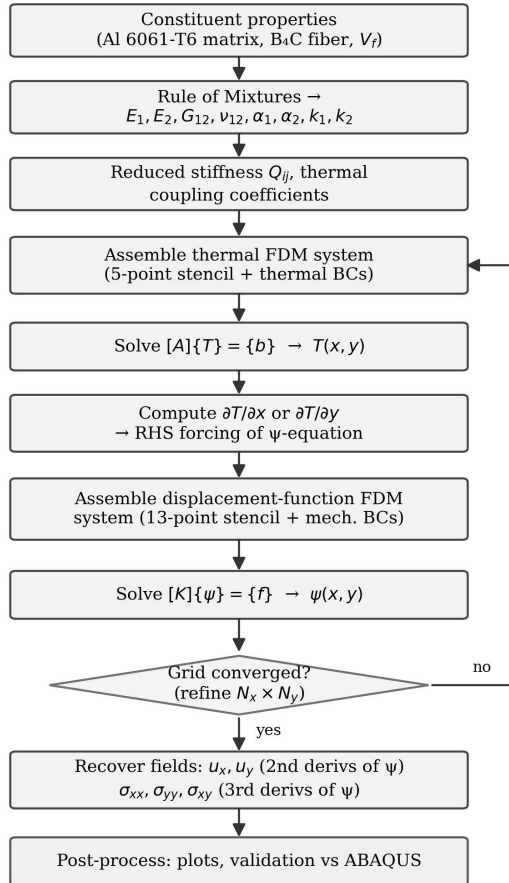


Figure 3.3: Solution flowchart for the coupled steady-state thermomechanical finite difference solver. The thermal system $[A]\{T\} = \{b\}$ is solved first to obtain $T(x, y)$; the temperature gradient then supplies the right-hand-side forcing of the displacement-function equation $[K]\{\psi\} = \{f\}$, from which the stress components σ_{xx} , σ_{yy} , τ_{xy} and displacement components u_x , u_y are recovered by differentiation of the converged ψ field.

3.7 Material Systems And Problem Specifications

3.7.1 Composite Material System

Both problems employ the Al–B₄C system: a 6061-T6 aluminium matrix reinforced with boron carbide fibers at a fiber volume fraction of $V_f = 0.50$, with fibers aligned along the x -axis ($\theta = 0^\circ$). The effective thermo-mechanical properties are listed in Table 3.1 and the reduced stiffness coefficients computed from Equation (3.12) are given in Equation (3.41).

Table 3.1: Effective thermo-mechanical properties of Al–B₄C at $V_f = 0.50$, estimated using the Rule of Mixtures.

Property	Symbol	Value
Longitudinal elastic modulus	E_1	234.5 GPa
Transverse elastic modulus	E_2	117.7 GPa
In-plane shear modulus	G_{12}	44.9 GPa
Major Poisson's ratio	ν_{12}	0.265
Minor Poisson's ratio	ν_{21}	0.133
Longitudinal CTE	α_1	$7.74 \times 10^{-6} \text{ K}^{-1}$
Transverse CTE	α_2	$16.6 \times 10^{-6} \text{ K}^{-1}$
Longitudinal thermal conductivity	k_1	$97 \text{ W m}^{-1} \text{ K}^{-1}$
Transverse thermal conductivity	k_2	$46.5 \text{ W m}^{-1} \text{ K}^{-1}$

$$Q_{11} = 240.0 \text{ GPa}, \quad Q_{22} = 120.2 \text{ GPa}, \quad Q_{12} = 31.9 \text{ GPa}, \quad Q_{66} = 44.9 \text{ GPa} \quad (3.41)$$

3.7.2 Problem I

The panel has in-plane dimensions $L_x = 3 \text{ m}$ and $L_y = 1 \text{ m}$ with uniform thickness $w = 0.01 \text{ m}$. A temperature difference is imposed in the y -direction; Problem I is therefore solved using Formulation II (Section 3.5.2). The boundary conditions are illustrated in Figure 3.4 and tabulated in Table 3.2.

3.7.3 Problem II

Problem II uses the same panel dimensions and fiber orientation as Problem I. The thermal gradient runs in the x -direction; Problem II is therefore solved using Formulation I

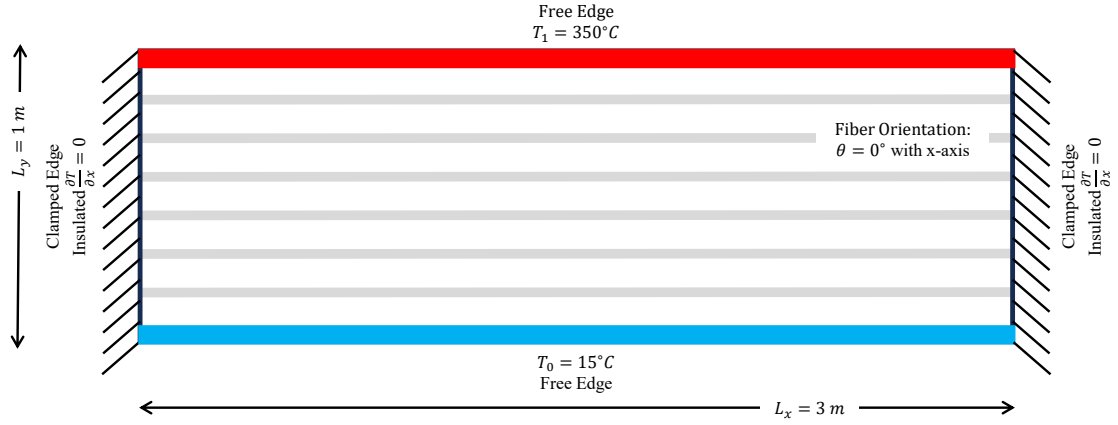


Figure 3.4: Problem I geometry and boundary conditions (y-directional thermal gradient, fully clamped lateral edges).

Table 3.2: Boundary conditions for Problem I ($T_1 = 350^\circ\text{C}$, $T_0 = 15^\circ\text{C}$, $\Delta T = 335^\circ\text{C}$).

Edge	Position	Thermal BC	Mechanical BC
Top	$y = L_y$	Isothermal, $T_1 = 350^\circ\text{C}$	Stress-free: $\sigma_{yy} = 0$, $\tau_{xy} = 0$
Bottom	$y = 0$	Isothermal, $T_0 = 15^\circ\text{C}$	Stress-free: $\sigma_{yy} = 0$, $\tau_{xy} = 0$
Left	$x = 0$	Insulated, $\partial T / \partial x = 0$	Clamped: $u_x = 0$, $u_y = 0$
Right	$x = L_x$	Insulated, $\partial T / \partial x = 0$	Clamped: $u_x = 0$, $u_y = 0$

(Section 3.5.1). The boundary conditions are illustrated in Figure 3.5 and tabulated in Table 3.3.

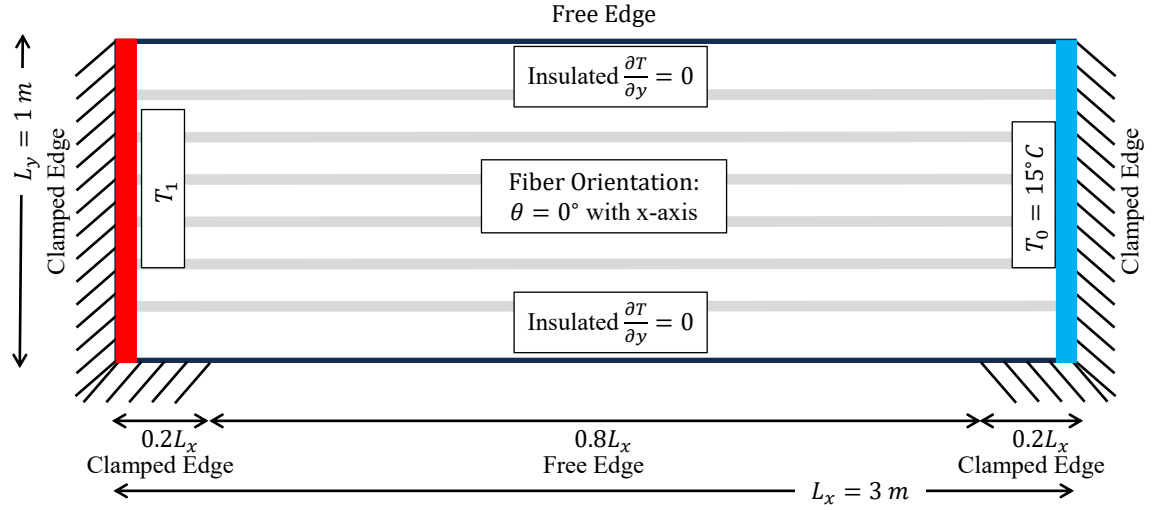


Figure 3.5: Problem II geometry and boundary conditions (x -directional thermal gradient, mixed bottom edge condition).

Table 3.3: Boundary conditions for Problem II ($T_0 = 15\text{ }^{\circ}\text{C}$, $T_{\text{ref}} = 0\text{ }^{\circ}\text{C}$).

Edge	Position	Thermal BC	Mechanical BC
Left	$x = 0$	Isothermal, T_1 {50, 100, 150, 300, 400} $^{\circ}\text{C}$	Roller: $u_x = 0$, $\tau_{xy} = 0$
Right	$x = L_x$	Isothermal, $T_0 = 15\text{ }^{\circ}\text{C}$	Roller: $u_x = 0$, $\tau_{xy} = 0$
Top	$y = L_y$	Insulated, $\partial T / \partial y = 0$	Stress-free: $\sigma_{yy} = 0$, $\tau_{xy} = 0$
Bottom	$y = 0$	Insulated, $\partial T / \partial y = 0$	Mixed: outer 20% at each end ($0 \leq x \leq 0.6\text{ m}$ and $2.4\text{ m} \leq x \leq L_x$) fully fixed ($u_x = 0$, $u_y = 0$); central 60% ($0.6\text{ m} < x < 2.4\text{ m}$) stress-free ($\sigma_{yy} = 0$, $\tau_{xy} = 0$)

The left-edge temperature T_1 is prescribed at five levels: 50, 100, 150, 300, and 400 $^{\circ}\text{C}$, while the right-edge temperature is held fixed at $T_0 = 15\text{ }^{\circ}\text{C}$ throughout.

3.8 Validation

The accuracy of the finite difference solver is established by a grid convergence study and cross-verification against ABAQUS finite element solutions for Problem I.

3.8.1 Grid Convergence Study

A grid refinement study is conducted for Problem I ($T_1 = 350^\circ\text{C}$, $T_0 = 15^\circ\text{C}$) to establish the minimum grid density required for grid-independent results. Six uniform grids of progressively increasing density are tested, and the peak values of σ_{xx} and σ_{yy} are extracted at the mid-section probe point ($x/L_x = 0.5$, $y/L_y = 0.5$) for each grid.

The convergence behaviour is shown in Figure 3.6. Both stress components exhibit a rapid rise from the coarsest grid (151×51) and progressively flatten as the grid is refined, approaching a well-defined asymptotic value. The relative change in peak σ_{xx} drops from 5.0% at the second grid to 0.3% at the 661×221 level, with only a 0.1% further change upon refinement to 751×251 . An identical trend is observed for σ_{yy} . The monotone decrease in successive changes is in consistent with the $\mathcal{O}(h^2)$ truncation error of the second-order central-difference stencils. On this basis, the 661×221 grid is selected as the production grid for all subsequent computations.

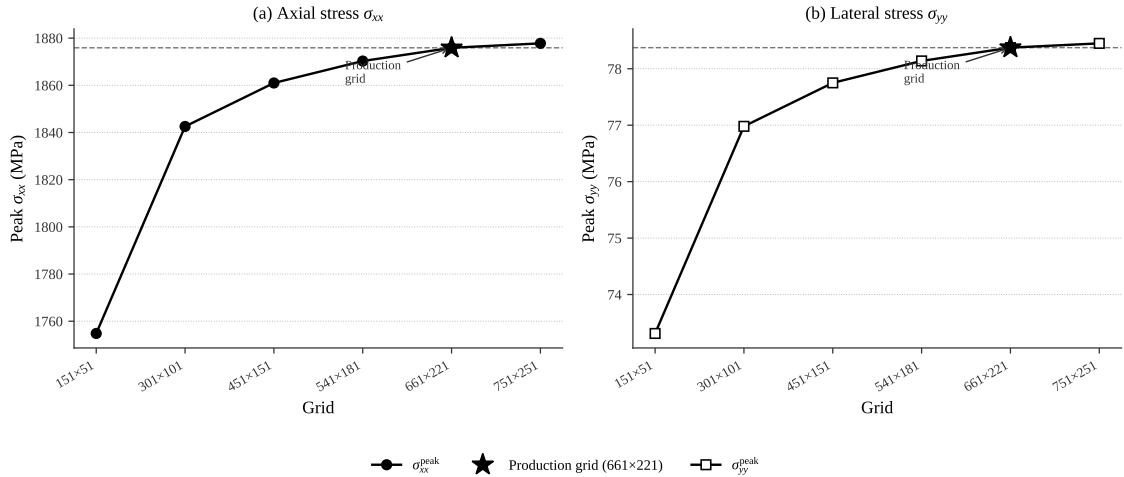


Figure 3.6: Convergence of peak σ_{xx} and σ_{yy} at the mid-section probe point ($x/L_x = 0.5$, $y/L_y = 0.5$) with increasing grid density, Problem I ($T_1 = 350^\circ\text{C}$, $T_0 = 15^\circ\text{C}$). The dashed line and star marker indicate the selected production grid (661×221 nodes).

3.8.2 Finite Element Reference Model

The finite element model uses four-node bilinear plane-stress elements (CPS4) with the same orthotropic material properties as listed in Table 3.1. The thermal boundary conditions of Problem I (Table 3.2) are applied as predefined temperature fields; the resulting temperature distribution supplies the thermal body force to the structural analysis via se-

quential coupling. The finite element mesh density is matched to the 661×221 production grid so that nodal comparisons are made at equivalent spatial locations.

Figure 3.7 shows the computed steady-state temperature distributions for both problems.

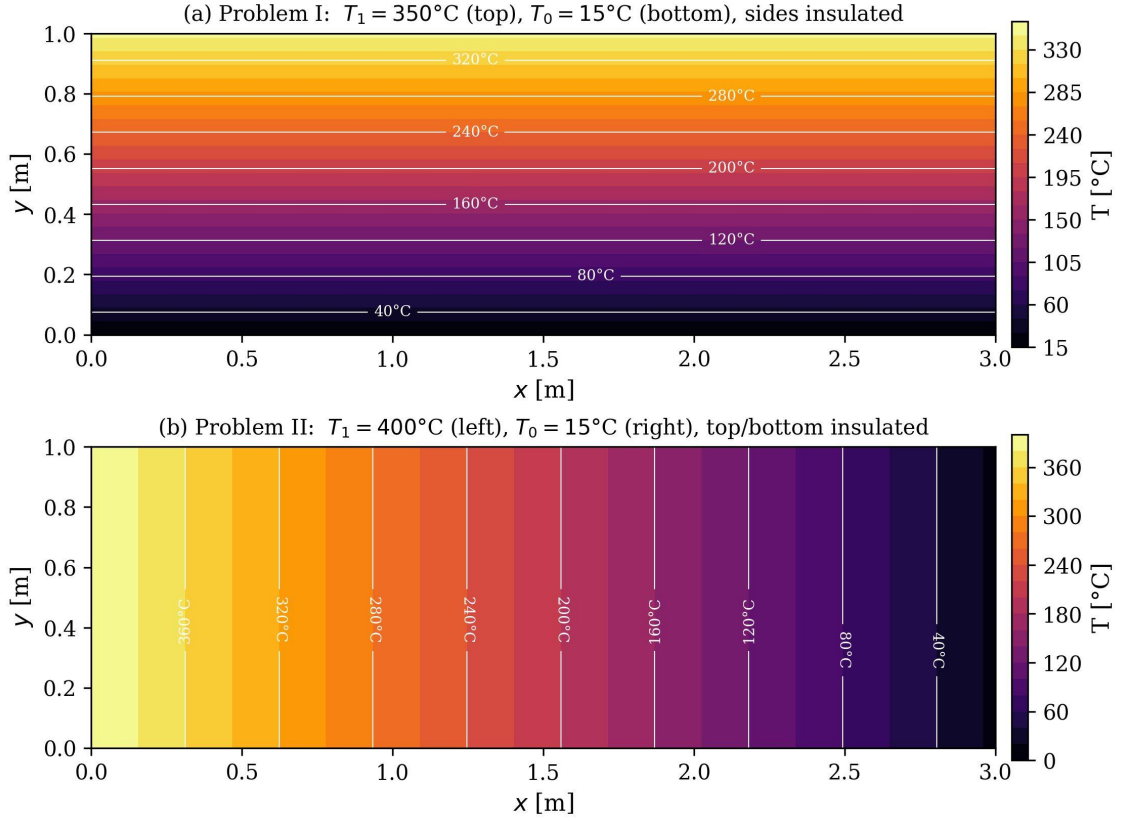


Figure 3.7: Steady-state temperature distributions $T(x, y)$ computed by the finite difference thermal solver. (a) Problem I: y -directional gradient with $T_1 = 350^\circ\text{C}$ at the top edge and $T_0 = 15^\circ\text{C}$ at the bottom. (b) Problem II: x -directional gradient with $T_1 = 400^\circ\text{C}$ at the left edge and $T_0 = 15^\circ\text{C}$ at the right.

3.8.3 Axial Stress Comparison

Figure 3.8(a) shows the axial stress distribution at the left clamped edge. Both methods predict a predominantly compressive axial stress across the plate width, in consistent with the thermal loading configuration. A notable discrepancy is observed at the top corner node ($y/L_y \approx 0$): the FDM solution produces a tensile spike of approximately $4.77 \times 10^9 \text{ Pa}$, whereas the ABAQUS solution gives a compressive value of approximately $-1.29 \times 10^9 \text{ Pa}$ at the same location. This divergence is attributed to the stress singularity at the intersection of the clamped mechanical condition and the free thermal

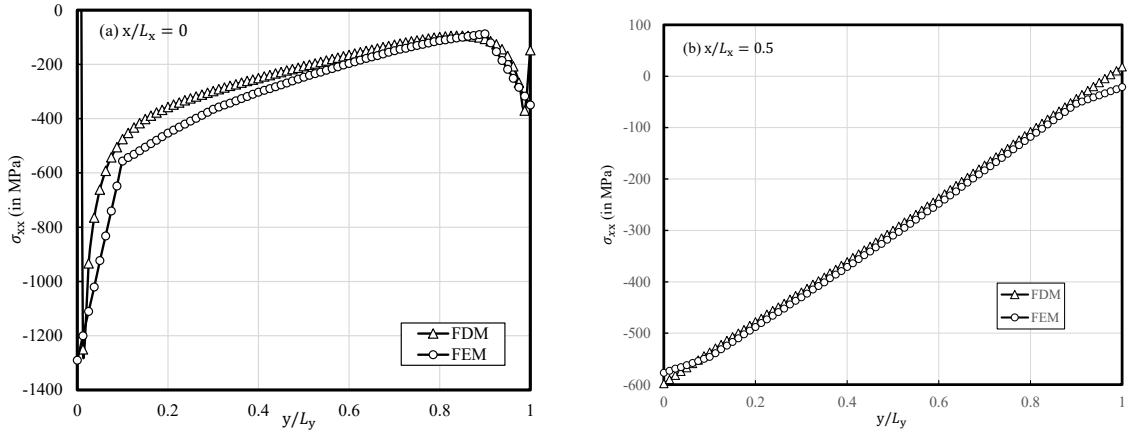


Figure 3.8: Axial stress (σ_{xx}) along the normalized plate width (y/L_y), Problem I: FDM vs. ABAQUS (FEM). (a) Left edge ($x/L_x = 0$); (b) mid-length section ($x/L_x = 0.5$).

edge, a well-recognised difficulty in numerical thermoelastic analysis (Noda et al., 2003; Timoshenko and Goodier, 1970). Away from the corner, the two solutions are found to converge progressively and agree well over the bulk of the plate width.

Figure 3.8(b) shows the axial stress at the mid-length section. The FDM and ABAQUS results are found to be in excellent agreement across the full plate width, with both exhibiting a nearly linear variation from approximately -6.0×10^8 Pa at $y/L_y = 0$ to near-zero at $y/L_y = 1$. This linear profile is in consistent with the linearly distributed temperature field between the isothermal top and bottom edges. The two curves are found to be virtually indistinguishable, confirming the validity of the FDM formulation at sections removed from the clamped lateral ends.

3.8.4 Lateral Stress Comparison

Figure 3.9(a) shows the lateral stress distribution at the left clamped edge. Both methods are found to predict a compressive lateral stress that decreases monotonically from a peak near $y/L_y \approx 0.025$ towards zero at the free horizontal edges. The FDM peak compressive value of approximately -8.0×10^8 Pa is slightly elevated relative to the ABAQUS value of approximately -7.75×10^8 Pa at the same location, a discrepancy found to be confined to the narrow corner zone adjacent to the clamped edge. For y/L_y between 0.1 and 0.9 the two solutions are found to agree well, with both displaying a near-linear compressive variation across the plate width.

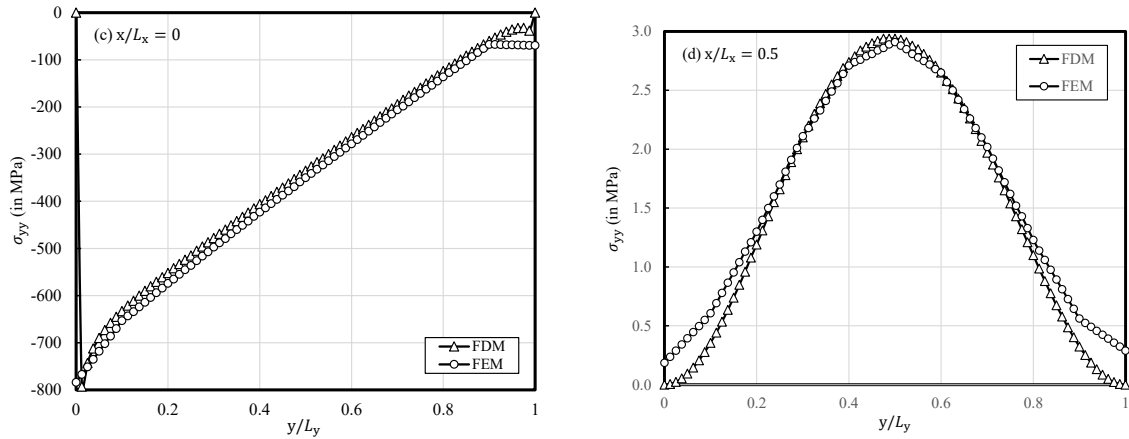


Figure 3.9: Lateral stress (σ_{yy}) along the normalized plate width (y/L_y), Problem I: FDM vs. ABAQUS (FEM). (a) Left edge ($x/L_x = 0$); (b) mid-length section ($x/L_x = 0.5$).

Figure 3.9(b) shows the lateral stress distribution at the mid-length section. Both methods are found to predict an arch-shaped distribution rising from near-zero at the free edges to a positive peak of 2.90×10^6 Pa (FDM) and 2.91×10^6 Pa (ABAQUS) near $y/L_y \approx 0.5$. The two curves are found to be in excellent agreement over the central region ($0.1 \leq y/L_y \leq 0.9$). Near the free edges the FEM solution retains a marginally higher residual lateral stress compared to the near-zero values produced by the FDM, which more closely satisfies the zero-stress free-edge condition.

The FDM stress results are in good to excellent agreement with the finite element benchmark across the bulk of the problem domain. Discrepancies are confined to the corner regions at the clamped lateral ends, where a stress singularity arises that neither numerical method resolves with full consistency (Noda et al., 2003; Timoshenko and Goodier, 1970). At the mid-length section, both σ_{xx} and σ_{yy} agree with the finite element reference to within the expected second-order accuracy. The validation establishes sufficient confidence in the solver to proceed with the parametric study of Chapter 4.

CHAPTER 4

RESULTS AND DISCUSSION

4.1 Overview

The thermoelastic stress results from the finite difference solutions of Problem I and Problem II are presented and discussed in two parts. The Problem I results constitute a parametric study, covering the influence of reinforcement material and fiber volume fraction on the thermoelastic stress field. The Problem II results examine the stress field of the Al–B₄C MMC plate under an x -directional thermal gradient, with attention to the effects of applied temperature differential and plate aspect ratio.

4.2 Problem I: Parametric Study

4.2.1 Effect Of Reinforcement Material

The influence of the reinforcement material on the thermoelastic stress field is investigated by comparing four aluminium-based metal matrix composite systems—Al–Graphite, Al–SiC, Al–B₄C, and Al–Al₂O₃—at a fixed fibre volume fraction of $V_f = 0.50$ under the Problem I boundary conditions (Section 3.7.2). The distributions of axial stress σ_{xx} and lateral stress σ_{yy} along the normalized plate length (x/L_x) at the mid-depth section ($y/L_y = 0.5$) are presented in Figure 4.1 for all four material systems.

The axial stress distributions obtained for the four composite systems are presented in Figure 4.1(a). The axial stress is found to be wholly compressive over the entire plate length for all four material systems, which is in accordance with the restraining action of the clamped lateral ends on the thermally induced elongation of the heated plate, and the distributions are observed to exhibit a characteristic non-linear profile with a pronounced compressive peak at an intermediate location along the plate length. The magnitude of the compressive stress is found to be governed by the effective longitudinal coefficient of thermal expansion of the laminate (Gibson, 2016; Kaw, 2005): the Al–Graphite system exhibits the smallest stress on account of the near-zero, slightly negative coefficient of thermal expansion of the graphite fibre, whereas the Al–Al₂O₃ system, possessing the highest fibre expansivity among the four systems considered, is found to develop the most

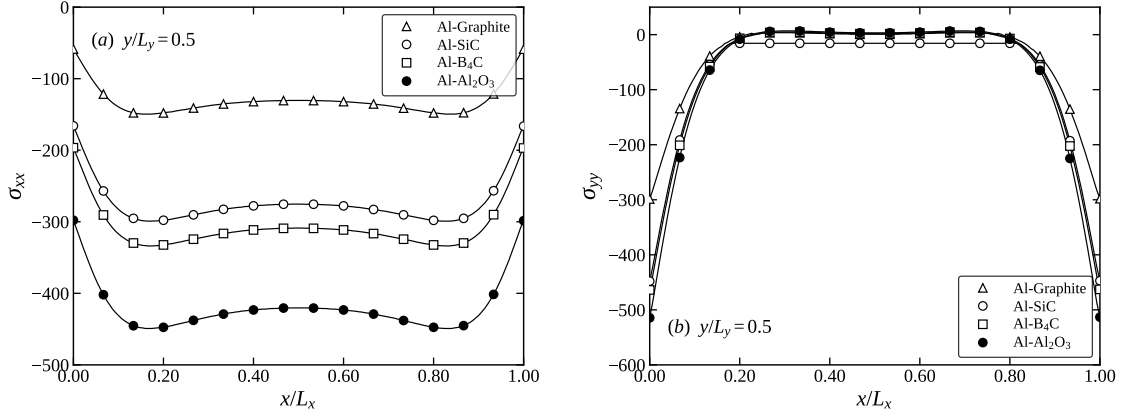


Figure 4.1: Distributions of (a) axial stress (σ_{xx}) and (b) lateral stress (σ_{yy}) along the normalized plate length (x/L_x) at the mid-height section ($y/L_y = 0.5$) for four aluminum-based MMC systems ($V_f = 0.50$, Problem I).

severe compression, approaching approximately -450 MPa at the peak location. The intermediate behaviour of the Al-SiC and Al-B₄C systems is, as expected, found to be in consistent with the corresponding magnitudes of their effective thermal expansion coefficients, thereby establishing a monotonic ordering of the axial stress magnitudes in the sequence Al-Graphite < Al-SiC $\approx$ Al-B₄C < Al-Al₂O₃.

The corresponding lateral stress distributions are presented in Figure 4.1(b). The lateral stress is found to be negligibly small over the interior of the plate for all four material systems, which is in consistent with the traction-free condition imposed on the upper and lower surfaces, and the four distributions are observed to be almost identical in terms of the magnitude and nature of variation throughout the central region of the plate length. A pronounced compressive intensification is, however, evident in the immediate vicinity of the clamped ends ($x/L_x = 0$ and $x/L_x = 1$), where the biaxial restraint of the support gives rise to a localised transverse stress reaction, with peak magnitudes reaching approximately -500 MPa for the Al-Al₂O₃ system. It is noted here that the slight departure of the Al-Graphite system at the end regions is attributed to its distinct thermal expansion characteristic, the influence of which is found to be confined to the regions of transition of physical conditions. The near-coincidence of all four curves over the interior thereby establishes that the lateral stress at the mid-depth section is predominantly governed by the boundary constraints rather than by the specific reinforcement material, and the present displacement-function formulation is found to capture this material-invariant interior behaviour with equal sophistication across all four systems.

4.2.2 Effect Of Fibre Volume Fraction

The influence of fibre volume fraction on the thermoelastic stress field is investigated for the Al–B₄C system by varying V_f from 0.3 to 0.7 in increments of 0.1, while retaining the Problem I boundary conditions and the plate geometry of $L_x/L_y = 3$. The distributions of axial stress σ_{xx} and lateral stress σ_{yy} along the normalized plate length (x/L_x) at the mid-depth section ($y/L_y = 0.5$) for the five volume fractions are presented in Figure 4.2.

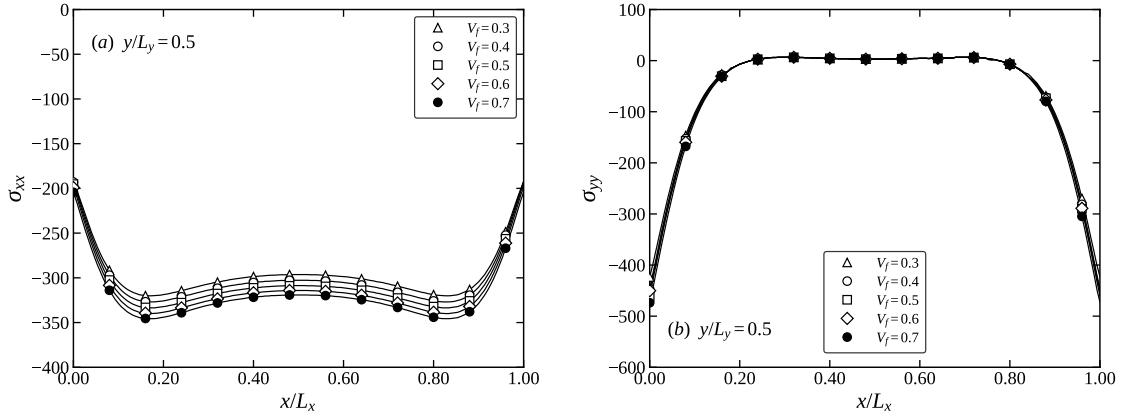


Figure 4.2: Distributions of (a) axial stress (σ_{xx}) and (b) lateral stress (σ_{yy}) along the normalized plate length (x/L_x) at the mid-height section ($y/L_y = 0.5$) for the Al–B₄C MMC system at five fiber volume fractions ($V_f = 0.3$ – 0.7 , Problem I).

The influence of the fibre volume fraction on the axial stress distribution of the Al–B₄C system is presented in Figure 4.2(a). The axial stress is found to remain wholly compressive for all five volume fractions considered, and the magnitude of the compressive stress is observed to increase progressively as the fibre content is raised from $V_f = 0.3$ to $V_f = 0.7$, the peak value shifting from approximately -300 MPa at $V_f = 0.3$ to approximately -345 MPa at $V_f = 0.7$. This monotonic trend is attributed to the dominant stiffening of the laminate with increasing reinforcement content, whereby the gain in longitudinal stiffness is found to outweigh the accompanying reduction in effective thermal expansion, thereby establishing a net intensification of the constrained axial stress. The distributions are found to retain a self-similar nature over the whole range of volume fractions, the curves being progressively displaced towards higher compression with increasing fibre content without any change in the qualitative character of the profile. It would be worth mentioning here that the non-linear peak in σ_{xx} is found to occur at a consistent normalized location of approximately $x/L_x \approx 0.15$ – 0.20 for all five cases, thereby indicating that the fibre volume fraction does not alter the position of the stress concentration along the plate length under the present boundary conditions.

The corresponding variation of the lateral stress with fibre volume fraction is presented in Figure 4.2(b). The lateral stress is found to be essentially zero over the interior of the plate for all five volume fractions, and the five distributions are observed to be virtually coincident throughout, thereby indicating that the transverse response at the mid-depth section is practically insensitive to the fibre content for the present case. The pronounced compressive concentrations at the clamped ends are found to persist for all volume fractions with only marginal variation in magnitude, the peak lateral stress at $x/L_x = 0$ being approximately -460 MPa across all five curves. It is noted here that the near-complete collapse of all five lateral stress distributions onto a single curve is a direct consequence of the traction-free interior condition and the dominance of the boundary-imposed biaxial restraint over the material stiffness variation in the transverse direction, which, in turn, establishes that σ_{yy} at the mid-depth section is governed predominantly by the mechanical boundary conditions rather than by the fibre volume fraction for the present Al-B₄C composite system.

4.3 Problem II: Stress Field Analysis

4.3.1 Effect Of Applied Temperature Differential

The distributions of lateral stress (σ_{yy}) along the normalized plate width (y/L_y) at the mid-length section ($x/L_x = 0.5$) are presented in the left panel of Figure 4.3 for all five temperature levels.

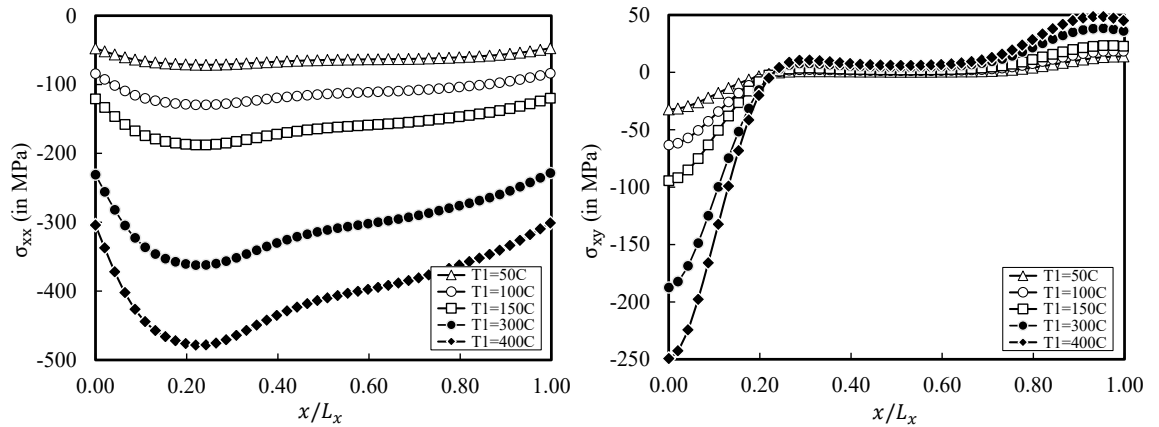


Figure 4.3: (a) Lateral stress (σ_{xx}) and (b) shear stress (σ_{xy}) along the normalized plate width (y/L_y) at the mid-length section ($x/L_x = 0.5$), Al-B₄C MMC plate ($L_x/L_y = 3$, Problem II) for five prescribed left-end temperatures T_1 .

The lateral stress at the mid-length section is found to be entirely compressive in nature throughout the plate width for all five temperature levels considered. The distributions are found to exhibit a pronounced concave or U-shaped profile, with σ_{yy} approaching a near-zero value at the top surface ($y/L_y \approx 1$) and attaining a maximum compressive magnitude in the lower portion of the plate width, approximately at $y/L_y \approx 0.10$ – 0.15 . The σ_{yy} minimum is found to be approximately -71 MPa for $T_1 = 50$ °C, increasing progressively in magnitude to approximately -479 MPa for $T_1 = 400$ °C, the increase being found to be proportional to the applied temperature differential and thereby in consistent with the principle of linear thermoelastic superposition (Noda et al., 2003).

The compressive nature of σ_{yy} at the mid-length section is attributable to the Poisson coupling between the dominant compressive axial stress (σ_{xx}) and the lateral response of the plate in the y -direction (Noda et al., 2003; Timoshenko and Goodier, 1970). As the x -directional temperature gradient induces a large compressive axial deformation throughout the plate, the corresponding Poisson expansion in the transverse y -direction is resisted by the structural continuity of the plate and, more critically, by the partial mechanical constraint imposed by the fixed segments of the bottom boundary. This constraint prevents the free transverse expansion that would otherwise be expected at the bottom region of the plate width, thereby generating the compressive lateral stress observed at the mid-length section. The characteristic U-shaped profile, with the stress minimum near $y/L_y \approx 0.10$ – 0.15 and a near-zero value at the top surface, is a direct reflection of the partially fixed bottom boundary condition and the free top surface acting simultaneously on the thermoelastic response of the plate.

The five curves are found to maintain a consistent relative ordering throughout the entire plate width without any intersection, which is in complete agreement with the principle of linear superposition applicable to the present thermoelastic problem, thereby establishing the proportional scaling of the lateral stress field with the applied temperature differential.

The distributions of shear stress (σ_{xy}) at the mid-length section are presented in the right panel of Figure 4.3 for all five temperature levels.

The shear stress distribution at the mid-length section is found to exhibit a sign-changing character across the plate width. It is noted here that for a plate with uniformly free top and bottom surfaces, no shear stress would be expected at the mid-length section under a one-dimensional x -directional thermal loading; the non-trivial shear stress field observed is therefore entirely attributable to the mixed and changeable mode of physical conditions at

the bottom surface, thereby underscoring the structural significance of the partial bottom-edge constraint in the present problem.

In the vicinity of the bottom surface ($0 \leq y/L_y \lesssim 0.15$), the shear stress is found to be strongly negative, with the peak compressive value occurring at the bottom surface ($y/L_y = 0$) itself, reaching approximately -32 MPa for $T_1 = 50^\circ\text{C}$ and escalating to approximately -249 MPa for $T_1 = 400^\circ\text{C}$. This large compressive shear stress concentration is attributed to the mechanical restraint imposed by the fixed segments of the bottom boundary, which resists the differential axial displacement that the x -directional temperature gradient seeks to impose at the bottom edge.

As the position increases from the bottom surface toward the upper region, the shear stress is found to transition from its large compressive value to a near-zero magnitude at approximately $y/L_y \approx 0.15$ – 0.20 , and subsequently to a small positive tensile value that is maintained at a nearly constant positive magnitude throughout the upper portion of the plate width up to the top surface. The positive shear stress in the upper region is found to increase with the prescribed temperature level from approximately $+14$ MPa for $T_1 = 50^\circ\text{C}$ to approximately $+49$ MPa for $T_1 = 400^\circ\text{C}$. The near-zero value of σ_{xy} at the top surface is in conformity with the prescribed free-surface mechanical boundary condition ($\sigma_{xy} = 0$ at $y/L_y = 1$).

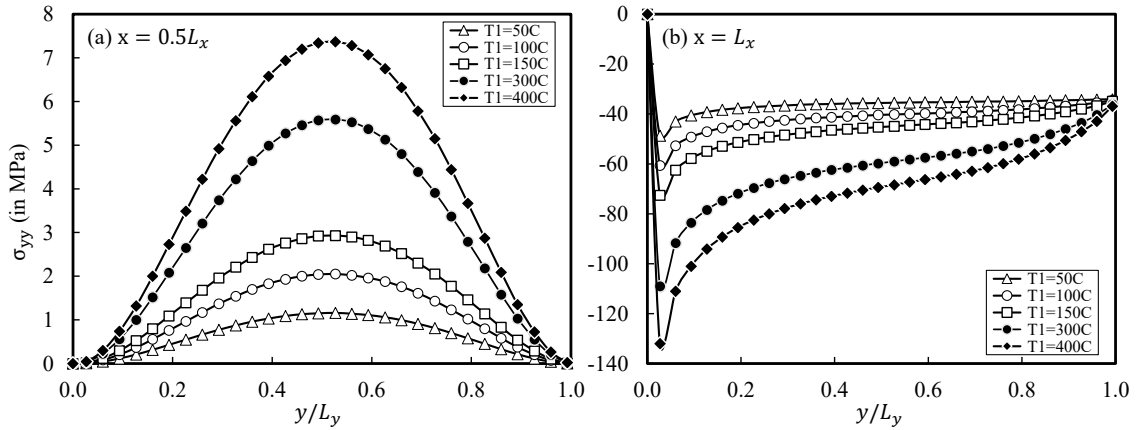


Figure 4.4: (a) Axial stress (σ_{xx}) along the normalized plate width (y/L_y) at the mid-length section ($x/L_x = 0.5$); (b) lateral stress (σ_{yy}) at the cold fixed end ($x/L_x = 1$). Al-B₄C MMC plate ($L_x/L_y = 3$, Problem II) for five prescribed left-end temperatures T_1 .

The distributions of σ_{yy} along the normalized plate width at the cold fixed end ($x/L_x = 1$) are presented in the right panel of Figure 4.4. The lateral stress at the cold fixed end is found to be entirely compressive throughout the plate width for all five temperature lev-

els. The distributions exhibit a non-linear profile, with σ_{yy} being approximately zero at both the top surface ($y/L_y = 1$) and the bottom surface ($y/L_y = 0$), and developing its maximum compressive magnitude in the interior region of the plate width at approximately $y/L_y \approx 0.05$ – 0.10 . The maximum compressive σ_{yy} at this section is found to be approximately -69 MPa for $T_1 = 50^\circ\text{C}$ and increases proportionally to approximately -194 MPa for $T_1 = 400^\circ\text{C}$.

The zero lateral stress at both the top and bottom surfaces is in conformity with the prescribed mechanical boundary conditions: the top surface is free ($\sigma_{yy} = 0$ at $y/L_y = 1$), while the zero lateral stress at the bottom corner is consistent with the corner modelling conditions at the junction of the fixed lateral end and the partially constrained bottom surface.

The physical mechanism responsible for the compressive lateral stress at the cold fixed end is the Poisson coupling transmitted through the plate from the thermally loaded interior. The x -directional temperature gradient drives a large compressive axial stress throughout the plate body; the associated Poisson expansion in the y -direction is resisted at the fixed lateral end by the displacement constraint $u_y = 0$, which prevents the free transverse expansion that would otherwise occur. The lateral stress distributions for different temperature levels are found to be geometrically similar in profile, differing only in magnitude and scaling proportionally with the applied temperature differential $\Delta T = T_1 - T_0$, which is in consistent with the expected linear thermoelastic character of the present problem (Noda et al., 2003).

The distributions of σ_{xx} along the normalized plate width (y/L_y) at the mid-length section ($x/L_x = 0.5$) are presented in the left panel of Figure 4.4. The axial stress at the mid-length section is found to be entirely tensile throughout the plate width for all five temperature levels, which is in marked contrast with the compressive character of σ_{yy} observed at the same mid-length section. The distributions exhibit a smooth bell-shaped parabolic profile that rises from zero at both surfaces ($y/L_y = 0$ and $y/L_y = 1$), attaining a maximum tensile value near $y/L_y \approx 0.50$ – 0.60 . The maximum tensile σ_{xx} at the mid-length section is found to be approximately $+1.16$ MPa for $T_1 = 50^\circ\text{C}$ and increases to approximately $+7.37$ MPa for $T_1 = 400^\circ\text{C}$.

The tensile axial stress at the mid-length section is physically attributable to the Poisson coupling between the dominant compressive lateral stress (σ_{yy}) at the mid-length section and the axial response of the plate. The compressive σ_{yy} , which is induced by the re-

strained Poisson expansion of the plate, in turn induces a Poisson contraction in the axial x -direction that is partially resisted by the structural continuity of the plate in the longitudinal direction, thereby generating the small tensile axial stress observed.

It is noted that the magnitudes of the tensile σ_{xx} along the plate width at the mid-length section are found to be considerably smaller than the corresponding compressive σ_{yy} values at the same mid-length position, the ratio of the maximum σ_{xx} to the maximum compressive σ_{yy} being found to be of the order of 1.5–1.6% for all temperature levels. This large difference in magnitude is in consistent with the physical characteristic of the Al–B₄C composite system, in which the significantly higher longitudinal stiffness (E_1) compared to the transverse stiffness (E_2) ensures that the Poisson-coupled tensile axial stress is considerably smaller in magnitude than the dominant lateral thermoelastic response of the plate.

4.3.2 Effect Of Plate Aspect Ratio

The influence of plate aspect ratio ($AR = L_x/L_y$) on the thermoelastic stress field is investigated for four values: $AR = 1, 2, 3$, and 4 , corresponding to $L_x = 1, 2, 3$, and 4 m with $L_y = 1$ m fixed throughout. The left-edge temperature is held at $T_1 = 400^\circ\text{C}$ and the right edge at $T_0 = 15^\circ\text{C}$ for all cases, so that the thermal boundary condition is identical across all aspect ratios and the geometric effect on the thermoelastic response is isolated. The distributions of axial stress σ_{xx} and lateral stress σ_{yy} along the normalized plate length (x/L_x) at the mid-width section ($y/L_y = 0.5$) are presented in Figure 4.5 (panels (a) and (b) respectively) for all four aspect ratios.

The distributions of axial stress σ_{xx} along the normalized plate length at the mid-width section, as apparent from Figure 4.5(a), are found to be compressive throughout the plate length for all aspect ratios considered, which is in consistent with the compressive state of axial thermal stress expected for a plate subjected to a positive temperature gradient in the x -direction under the present mechanical boundary conditions.

The axial stress distributions are found to exhibit a characteristic non-linear profile with a pronounced compressive peak at an intermediate location along the plate length, the position of which is found to shift progressively toward $x/L_x = 0$ as the aspect ratio is increased. The peak compressive intensity in σ_{xx} is found to be of the order of approximately -489 to -500 MPa, and the magnitude and nature of variation of the distributions are found to change considerably with the aspect ratio, particularly in the region between

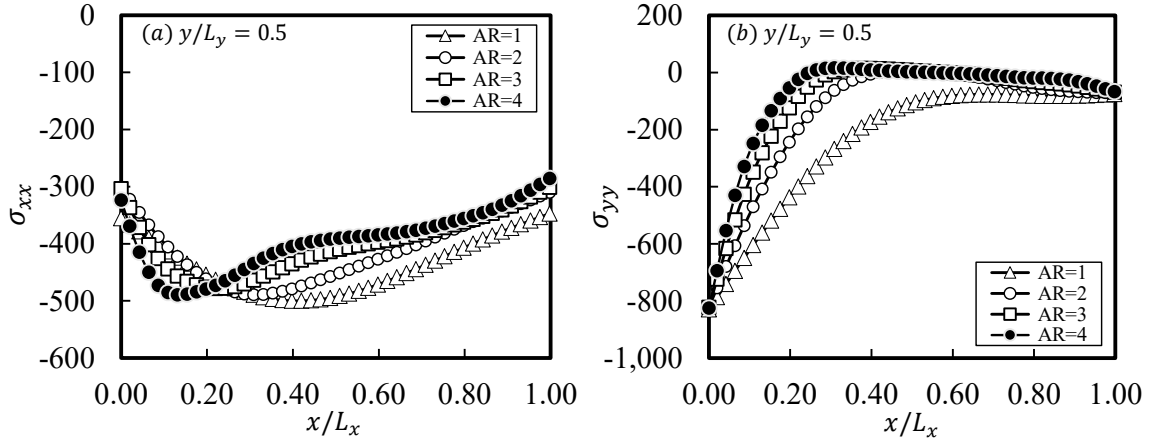


Figure 4.5: Distributions of (a) axial stress (σ_{xx}) and (b) lateral stress (σ_{yy}) along the normalized plate length (x/L_x) at the mid-height section ($y/L_y = 0.5$) for four plate aspect ratios ($T_1 = 400^\circ\text{C}$, $T_0 = 15^\circ\text{C}$, Problem II).

the mixed mechanical boundary at the bottom edge ($y/L_y = 0$) and the free top edge ($y/L_y = 1$). For $AR = 1$, the peak compressive axial stress is found to occur at approximately $x/L_x \approx 0.41$, while for $AR = 4$ this peak is found to shift to $x/L_x \approx 0.14$, indicating the significant influence of plate geometry on the distribution of σ_{xx} . It would be worth mentioning that, unlike the lateral stress distributions which show relatively minor sensitivity to aspect ratio except near the cold boundary, the axial stress distributions are found to be considerably more sensitive to the geometric change throughout the entire plate length. This greater sensitivity is primarily attributed to the dependence of the axial stress state on the two-dimensional thermoelastic equilibrium, which is governed in part by the plate aspect ratio through the displacement compatibility conditions.

The distributions of lateral stress σ_{yy} along the normalized plate length at the mid-width section, as apparent from Figure 4.5(b), are found to be compressive throughout the plate length for all aspect ratios considered, with the maximum compressive intensity occurring at the hot left boundary ($x/L_x = 0$) and decreasing monotonically toward the cold right boundary ($x/L_x = 1$). The maximum lateral stress at $x/L_x = 0$ is found to be of the order of approximately -825 MPa, and this peak value is observed to be relatively insensitive to the change in aspect ratio, as all four distributions are found to converge to nearly identical values at the hot boundary. This is in consistent with the physical interpretation that the extreme surface stress is predominantly governed by the prescribed temperature condition at that surface, rather than by the overall plate geometry.

A noticeable sensitivity of the lateral stress to the aspect ratio is, however, observed as one moves away from the hot boundary toward $x/L_x = 1$. The distributions for the

higher aspect ratios ($AR = 3$ and $AR = 4$) are found to exhibit a change in sign and to attain small tensile values in the vicinity of the cold boundary ($x/L_x \rightarrow 1$), which is not observed for $AR = 1$. This transition from a compressive to a tensile state of lateral stress at the cold boundary with increasing aspect ratio is found to be a consequence of the increasing restraint imposed by the longer plate in the x -direction on the thermally induced deformation, thereby generating a redistributed stress state that manifests tension at the cooler boundary for the more slender plate configurations.

The overall results from the present aspect ratio study reveal that the thermoelastic response of the composite plate is found to be non-trivially dependent on the plate geometry, especially in the regions of transition of physical conditions along the plate length. The present displacement-function modelling scheme is found to capture these geometric effects with equal accuracy across all aspect ratios considered, thereby further establishing the soundness and appropriateness of the present thermoelastic formulation for composite plate configurations of varying aspect ratios.

CHAPTER 5

CONCLUSIONS AND RECOMMENDATIONS

5.1 Overview

A from-first-principles finite difference framework for the coupled steady-state thermomechanical analysis of an orthotropic fiber-reinforced composite panel has been developed and applied to two distinct boundary value problems involving an Al–B₄C metal matrix composite plate. The governing equations for the orthotropic heat conduction problem and the plane-stress thermoelastic equilibrium problem were derived analytically from the respective constitutive relations and balance laws. A scalar displacement function $\psi(x, y)$ was introduced to reduce the coupled system to a single fourth-order partial differential equation, with two independent formulations corresponding to temperature gradients in each coordinate direction. Both governing equations were discretized on a uniform rectangular grid and solved numerically. The solver was validated against commercial finite element solutions, and a parametric stress study was conducted across two boundary value problems. The principal conclusions and recommendations for future work are presented below.

5.2 Conclusions

The following conclusions are drawn from the present investigation:

- (i) The steady-state orthotropic heat conduction equation with lumped surface convection is found to be amenable to a straightforward five-point central-difference discretization that correctly preserves the conductivity anisotropy ratio k_1/k_2 and satisfies the prescribed isothermal and insulated boundary conditions with second-order spatial accuracy.
- (ii) The displacement-function approach is found to successfully reduce the coupled plane-stress thermoelastic equilibrium system to a single fourth-order partial differential equation for the scalar potential $\psi(x, y)$. The two formulations (Formulation I for x -directional gradients, Formulation II for y -directional gradients) are found to share an identical left-hand operator, differing only in the thermal forcing

term on the right-hand side, which is found to confirm the internal consistency of the displacement-function method for the orthotropic thermoelastic problem class.

- (iii) The validation study for Problem I demonstrates that the present finite difference solver is found to produce axial stress (σ_{xx}) and lateral stress (σ_{yy}) distributions in good to excellent agreement with ABAQUS finite element solutions over the bulk of the problem domain. The discrepancies that are found to be present are confined to the corner regions at the clamped lateral ends, where the stress singularity at the intersection of the clamped mechanical condition and the free thermal edge is a well-recognised difficulty that affects both finite difference and finite element solutions in the absence of specialised corner treatment.
- (iv) The thermoelastic stress field of Problem II is found to be characterised by a dominant compressive lateral stress (σ_{yy}) of considerable magnitude at the mid-length section, a sign-changing shear stress (σ_{xy}) that is concentrated in the vicinity of the partially fixed bottom edge, and a small tensile axial stress (σ_{xx}) along the plate length at the mid-width section. The co-existence of these stress components is found to be a direct consequence of the mixed and changeable mechanical boundary conditions at the bounding surfaces of the present plate.
- (v) All computed stress components are found to scale proportionally with the applied temperature differential $\Delta T = T_1 - T_0$ across all five temperature levels investigated, which is found to be in consistent with the expected linear thermoelastic character of the present material system and boundary value problem formulation.
- (vi) The present finite difference framework is found to constitute a transparent, reproducible, and parameter-efficient computational scheme for the orthotropic thermoelastic problem class that is capable of simultaneously satisfying mixed and changeable boundary conditions with equal accuracy at each prescribed boundary segment.

5.3 Recommendations For Future Work

The following extensions of the present work are recommended for future investigation:

- (i) Extension of the present two-dimensional plane-stress framework to a three-dimensional thermoelastic formulation to capture through-thickness stress gradients and inter-laminar effects that are not accessible in the present model.

- (ii) Incorporation of transient thermal analysis to address the response of the composite plate under time-varying thermal loading and to assess the evolution of the thermoelastic stress field from the initial state to the steady state.
- (iii) Extension to off-axis fiber orientations ($\theta \neq 0^\circ$) using the general transformed stiffness formulation to investigate the influence of fiber angle on the thermoelastic field and to identify orientations that minimize peak stress.
- (iv) Application of failure criteria (Von Mises, Tsai-Wu, and maximum stress) to the computed stress fields to identify the conditions under which the composite plate is expected to approach the limits of its structural integrity under the thermal loading configurations studied.
- (v) Experimental validation of the computed thermoelastic stress fields through digital image correlation or strain gauge measurements on fabricated Al-B₄C composite specimens, to supplement the present computational cross-verification with commercial finite element solutions.
- (vi) Parametric investigation of the influence of fiber volume fraction on the thermoelastic response across a broader range of V_f values, with attention to the transition from matrix-dominated to fiber-dominated behaviour in the effective elastic and thermal properties.

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APPENDIX A

MATHEMATICAL DERIVATIONS

A.1 Overview

This appendix collects standard derivations whose results are used in the main text but whose step-by-step development is available in Kaw (2005) and Jones (1999). Cross-references from Chapter 3 points to the relevant sections below.

A.2 Plane-Stress Reduction

Starting from the full 3D orthotropic compliance relation (Kaw, 2005), the plane-stress conditions $\sigma_3 = \tau_{23} = \tau_{13} = 0$ yield the non-zero out-of-plane strain:

$$\varepsilon_3 = S_{13}\sigma_1 + S_{23}\sigma_2 \quad (\text{A.1})$$

This strain is excluded from the reduced in-plane system since it does not feed back into the plane-stress equilibrium equations; it may be recovered as a post-processing quantity. The remaining in-plane compliance relation is:

$$\begin{bmatrix} \varepsilon_1 \\ \varepsilon_2 \\ \gamma_{12} \end{bmatrix} = \begin{bmatrix} S_{11} & S_{12} & 0 \\ S_{12} & S_{22} & 0 \\ 0 & 0 & S_{66} \end{bmatrix} \begin{bmatrix} \sigma_1 \\ \sigma_2 \\ \tau_{12} \end{bmatrix} \quad (\text{A.2})$$

where the engineering-constant definitions of S_{ij} are:

$$S_{11} = \frac{1}{E_1}, \quad S_{22} = \frac{1}{E_2}, \quad S_{12} = -\frac{\nu_{12}}{E_1}, \quad S_{66} = \frac{1}{G_{12}} \quad (\text{A.3})$$

Inverting (A.2) and comparing with the reduced stiffness form gives the expressions for Q_{ij} stated in Eq. (3.12) of the main text.

A.3 Transformed Reduced Stiffness Coefficients

With $c = \cos \theta$ and $s = \sin \theta$, the transformed reduced stiffness coefficients are (Jones, 1999; Kaw, 2005):

$$\bar{Q}_{11} = Q_{11}c^4 + 2(Q_{12} + 2Q_{66})s^2c^2 + Q_{22}s^4 \quad (\text{A.4a})$$

$$\bar{Q}_{12} = (Q_{11} + Q_{22} - 4Q_{66})s^2c^2 + Q_{12}(s^4 + c^4) \quad (\text{A.4b})$$

$$\bar{Q}_{22} = Q_{11}s^4 + 2(Q_{12} + 2Q_{66})s^2c^2 + Q_{22}c^4 \quad (\text{A.4c})$$

$$\bar{Q}_{16} = (Q_{11} - Q_{12} - 2Q_{66})sc^3 - (Q_{22} - Q_{12} - 2Q_{66})s^3c \quad (\text{A.4d})$$

$$\bar{Q}_{26} = (Q_{11} - Q_{12} - 2Q_{66})s^3c - (Q_{22} - Q_{12} - 2Q_{66})sc^3 \quad (\text{A.4e})$$

$$\bar{Q}_{66} = (Q_{11} + Q_{22} - 2Q_{12} - 2Q_{66})s^2c^2 + Q_{66}(s^4 + c^4) \quad (\text{A.4f})$$

A.4 Transformed Reduced Compliance Coefficients

The inverse of (3.11) gives the transformed reduced compliance $\bar{S}_{ij}$ (Kaw, 2005):

$$\bar{S}_{11} = S_{11}c^4 + (2S_{12} + S_{66})s^2c^2 + S_{22}s^4 \quad (\text{A.5a})$$

$$\bar{S}_{12} = S_{12}(s^4 + c^4) + (S_{11} + S_{22} - S_{66})s^2c^2 \quad (\text{A.5b})$$

$$\bar{S}_{22} = S_{11}s^4 + (2S_{12} + S_{66})s^2c^2 + S_{22}c^4 \quad (\text{A.5c})$$

$$\bar{S}_{16} = (2S_{11} - 2S_{12} - S_{66})sc^3 - (2S_{22} - 2S_{12} - S_{66})s^3c \quad (\text{A.5d})$$

$$\bar{S}_{26} = (2S_{11} - 2S_{12} - S_{66})s^3c - (2S_{22} - 2S_{12} - S_{66})sc^3 \quad (\text{A.5e})$$

$$\bar{S}_{66} = 2(2S_{11} + 2S_{22} - 4S_{12} - S_{66})s^2c^2 + S_{66}(s^4 + c^4) \quad (\text{A.5f})$$

A.5 Stress And Strain Transformation

The stress components transform between material (1, 2) and global (x, y) axes via (Kaw, 2005):

$$\begin{bmatrix} \sigma_1 \\ \sigma_2 \\ \tau_{12} \end{bmatrix} = [T] \begin{bmatrix} \sigma_{xx} \\ \sigma_{yy} \\ \tau_{xy} \end{bmatrix}, \quad [T] = \begin{bmatrix} c^2 & s^2 & 2sc \\ s^2 & c^2 & -2sc \\ -sc & sc & c^2 - s^2 \end{bmatrix} \quad (\text{A.6})$$

The strain transformation uses the Reuter matrix $[R]$ to account for the engineering shear strain convention:

$$\begin{bmatrix} \varepsilon_1 \\ \varepsilon_2 \\ \gamma_{12} \end{bmatrix} = [T][R] \begin{bmatrix} \varepsilon_{xx} \\ \varepsilon_{yy} \\ \gamma_{xy} \end{bmatrix} [R]^{-1}, \quad [R] = \begin{bmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 2 \end{bmatrix} \quad (\text{A.7})$$

A.6 Energy Balance For The Orthotropic Panel

Consider a control volume $\Delta x \Delta y \Delta z$ within the panel. The net heat flux entering through the x - and y -faces, combined with convective loss through the top and bottom surfaces and a volumetric source g , must equal the rate of energy storage:

$$\begin{aligned} & -\frac{\partial Q_x}{\partial x} \Delta x \Delta y \Delta z - \frac{\partial Q_y}{\partial y} \Delta x \Delta y \Delta z \\ & - 2h_c(T - T_0)\Delta x \Delta y + g \Delta x \Delta y \Delta z = \rho C_p \Delta x \Delta y \Delta z \frac{\partial T}{\partial t} \end{aligned} \quad (\text{A.8})$$

The factor of 2 in the convective term accounts for heat loss from both the top ($z = \Delta z$) and bottom ($z = 0$) surfaces. Dividing through by $\Delta x \Delta y \Delta z$:

$$-\frac{\partial Q_x}{\partial x} - \frac{\partial Q_y}{\partial y} - \frac{2h_c}{\Delta z}(T - T_0) + g = \rho C_p \frac{\partial T}{\partial t} \quad (\text{A.9})$$

Substituting the Fourier flux definitions (3.5) and setting $\partial T / \partial t = 0$ with constant k_1, k_2 gives the steady-state governing equation (3.6).

It is further noted that the identical form of the left-hand operator in Formulation I and Formulation II means that the validation of the solver architecture for Problem I provides structural evidence for the correctness of the implementation used in Problem II as well; the only difference between the two problems is the right-hand side forcing term, which does not affect the operator structure.